

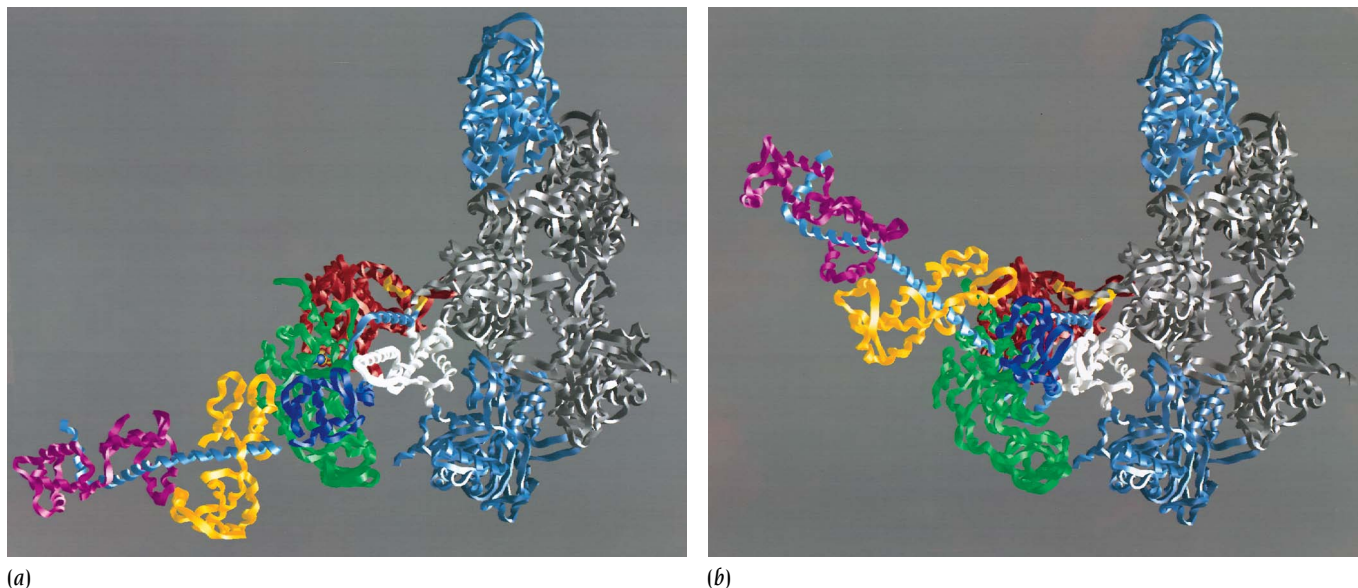


Figure 14.16 Structure of the actin-myosin complex, in the absence of bound nucleotide (a) and with ADP-vanadate bound (b). Five actin subunits (two pale blue and three gray) are shown on the right of each panel. The myosin on the left of each panel is colored similarly to Figure 14.15 with the light chains yellow and magenta and the 25-kDa domain green. The 50-kDa domain is colored red and white to show the 50K upper and 50K lower domains, respectively; a portion of the 20-kDa fragment, containing the converter region, is dark blue; a short linker region between the 20-kDa and the 50-kDa domains is shown in yellow; and the first part of the 20-kDa domain that is proposed to form part of the “hinge” region of the myosin head is shown in pale blue between the red, green and white domains. Note that the view is rotated with respect to Figure 14.15. (Courtesy of Ken Holmes.)

The groups of Rayment and Holmes fitted the atomic structures of F-actin and the S1 fragment of myosin into a low-resolution three-dimensional cryoelectron microscopy image of the actin-myosin complex determined by Ronald Milligan, Scripps Research Institute, California and obtained a detailed model of the complex (Figure 14.16a). This model establishes the spatial orientation of the S1 myosin fragment and shows that a cleft in myosin extends from the ATP-binding site to the actin-binding site. These sites are likely to be coupled by the opening and closing of this cleft. Furthermore, the very extended C-terminal α -helical tail of S1 is ideally placed to be a lever arm. The lever arm joins onto the bulk of the molecule via a small compact “converter domain” that lies just distal to a kinked α helix containing two reactive thiol groups. Numerous experiments point to the putative “hinge” for the lever arm being in the region of these thiol groups.

The structure of myosin supports the swinging cross-bridge hypothesis

According to the swinging cross-bridge scheme (see Figure 14.12) the myosin cross-bridge is envisaged as having two discrete conformations: (1) when it first attaches to actin with the ADP still bound and with the lever at the beginning of the working stroke; and (2) at the end of the working stroke when the ADP is released. The switch from state 1 to state 2 is often referred to as the “power stroke.” The end state is referred to as “rigor,” since it is the state muscles enter on ATP depletion when they become locked in *rigor mortis*. It is also called “strong” because it binds to actin quite tightly. The initial state is called the “weak binding state” because of its low affinity for actin. We might anticipate that these two states of the myosin cross-bridge exist independently from actin and indeed protein crystallography shows this to be the case.

The crystal structure of chicken S1, which was solved without bound nucleotide, fits well into the electron micrograph reconstructions of the strong actin-myosin, nucleotide-free interaction. In addition, Rayment and his colleagues at the University of Wisconsin have studied a crystalline truncated fragment of the *Dictyostelium* myosin II cross-bridge. The truncation eliminates the lever arm and the associated light chains, helping crystallization. However, the converter domain is still present. The crystal structures of this fragment have been determined with a number of ATP analogs bound, particularly ADP-vanadate, which is considered to be an analog of the transition state or possibly of the ADP-P_i state. The ADP-vanadate structure

shows dramatic changes in the shape of S1 compared with the chicken structure. There is a closing of the cleft associated with a movement of the helix corresponding to the switch II region in G-proteins (see Chapter 13), which is coupled to a rotation of the converter domain by 70° .

A model of this new state is shown in Figure 14.16b and compared with the chicken S1 structure. The coordinates of the missing lever arm have been generated from the chicken coordinates by superimposing the converter domains. Since the converter domain has been rotated 70° the end of the lever arm has moved through $120 \text{ \AA}$ along the actin helix axis, which is greater than most estimates of the size of the power stroke. The mechanism for coupling the movement of the lever arm with the status of the nucleotide-binding pocket that is revealed by this structure suggests that the two events are tightly coupled: cleft closed, lever up (beginning); cleft open, lever down (end).

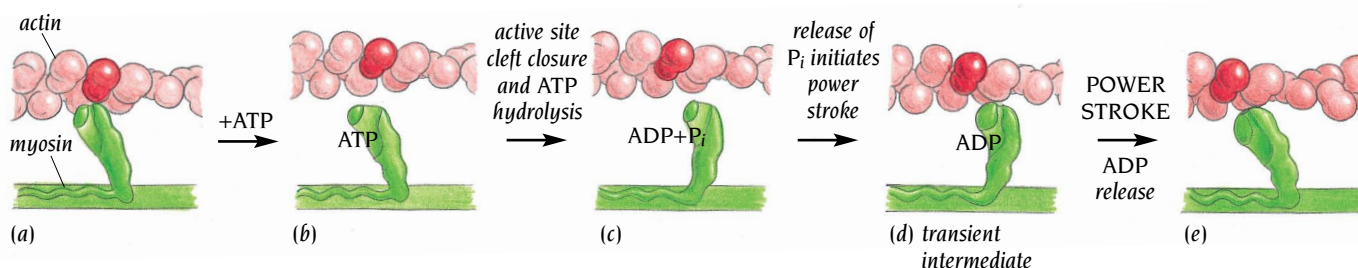
The role of ATP in muscular contraction has parallels to the role of GTP in G-protein activation

The essence of the sliding filament model is that the myosin head binds to the actin filament in one position relative to its anchor point on the myosin filament, changes this relative position about $100 \text{ \AA}$ along the fiber axis, and during this process the two filaments slide relative to each other by about the same distance. The myosin head then detaches from the actin filament ready to repeat the process. We will now discuss the molecular mechanisms of the power stroke suggested by Ken Holmes, from the known structures of myosin and the actin–myosin complex, that couple ATP binding and hydrolysis, myosin–actin interactions and the repositioning of the myosin heads by conformational changes.

In the absence of nucleotides, the myosin nucleotide-binding cleft is open, the lever arm is “down,” the actin-binding site is intact and this form binds strongly to actin (Figure 14.17a). This is the rigor state into which in the absence of nucleotides the muscle is locked as in *rigor mortis*. If ATP is added, the myosin head, bound to actin, will bind ATP, and then dissociate from the actin (Figure 14.17b). Binding of ATP to the nucleotide-binding domain in the cleft causes the P loop, which corresponds to the switch II region in G-proteins, described in Chapter 13, to change its conformation. The γ -phosphate of ATP plays the same role in this conformational change as the γ -phosphate of GTP in G-proteins. This change in the loop conformation is coupled to a major conformational change of parts of the head protein; as a result the cleft closes and the region that binds actin releases the actin filament. The bound ATP is then hydrolyzed to ADP and phosphate (Figure 14.17c).

It seems likely, therefore, that as the bound phosphate molecule is released, the cleft starts to open and the myosin head binds to actin (Figure 14.17d). Release of ADP coincides with a conformational change that fully opens the myosin cleft, causing actin to be tightly bound, and moves the lever arm to the “up” position. Since the myosin head is now strongly bound to actin at one end and covalently linked to the myosin fibril at the other

Figure 14.17 A sequence of events combining the swinging cross-bridge model of actin and myosin filament sliding with structural data of myosin with and without bound nucleotides. (a) Myosin in the absence of nucleotide binds tightly to actin in a 45° conformation. (b) The binding of ATP induces myosin to dissociate from actin. (c) As a result of ATP hydrolysis the myosin undergoes a conformational switch to the 90° state. (d) The release of phosphate coincides with myosin binding weakly to actin, and the subsequent release of ADP returns the myosin to the initial 45° state. This conformational switch completes the power stroke that drives the sliding of the actin and myosin filaments relative to each other by about $100 \text{ \AA}$. (Adapted from B. Alberts et al, *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.)



end, the conformational change leads to a relative displacement of the actin and myosin filaments; they slide past each other (Figure 14.17e).

The conformational change of myosin that leads to the sliding of the filaments thus consists of switching a lever arm that changes the relative position of the two ends of the elongated heads by about 100 Å. Part of the energy of ATP hydrolysis must be stored as a mechanical “spring” to drive this conformational change. The two major states described here are not the only states of the actin–myosin complex; additional intermediate states are now being defined but it is unlikely that we shall have to change the current overall picture of the mechanism of muscle contraction.

Conclusion

Fibrous proteins are long-chain polymers that are used as structural materials. Most contain specific repetitive amino acid sequences and fall into one of three groups: coiled-coil α helices as in keratin and myosin; triple helices as in collagen; and β sheets as in silk and amyloid fibrils.

The coiled-coil fibrous proteins have heptad repeats in their amino acid sequence and form oligomers—usually dimers or trimers—through their coiled coils. These oligomeric units then assemble into fibers.

Collagen is a large trimeric superhelical molecule about 3000 Å long. Each of the three polypeptide chains has about 1000 amino acids, and is folded into an extended left-handed helix with a rise per turn of 9.6 Å. Proper assembly of the chains requires propeptide regions that are cleaved upon maturation of the triple helix. The three helical subunits coil about a central axis to form a right-handed superhelix with a 100-Å repeat distance. Steric constraints within this superhelix account for the large number of repeat sequences Gly-X-Y, where X is proline and Y is hydroxyproline, in the polypeptide chains. The crystal structure of a synthetic collagen-like peptide has shown the importance of both direct and water-mediated hydrogen bonds in stabilizing the triple helix.

Amyloid fibers, which are deposited in cells of the eye, brain, and heart, amongst others, are associated with a variety of inherited or infectious diseases. The amyloid fibers are formed as a result of the refolding of normally globular soluble proteins, which leads to their polymerization into long, straight, unbranched fibrils, about 100 Å in diameter, that aggregate into fibers. In a rare and fatal disease, familial amyloidotic polyneuropathy, point mutations in transthyretin, a carrier of the hormone thyroxine, causes the normally soluble globular transthyretin to deposit in the heart and brain. The structure of normal transthyretin and the fibrillar form have been studied by high-resolution crystallography and fiber diffraction, respectively. A model for the fibrillar form has been proposed, with four parallel protofilaments packed together to form a fibril. Each protofilament is built up from four continuous twisted β sheets running parallel to the fiber axis, with the constituent β strands in each at right angles to the axis.

Silk fibers, which have incredible strength, comprise well-ordered microcrystals of β -sheets that make up about 30% of the protein mass, interspersed in a matrix of polypeptide chains without order. The β strands of the sheets are oriented parallel to the fiber axis.

Muscle is composed of two sets of interdigitating filaments: thick filaments containing the fibrous protein myosin and thin filaments containing F-actin, a helical polymer of the soluble, globular, monomeric G-actin. In addition there are a number of other muscle proteins that regulate muscle activity. When muscles contract, actin filaments and myosin filaments slide with respect to each other. This movement is brought about by rounds of conformational changes of the heads of myosin molecules that project as cross-bridges from the myosin thick filaments to actin subunits in F-actin. Each round requires the hydrolysis of one molecule of ATP. Detailed structural studies of fragments of myosin and of actin at atomic resolution have clarified many of the molecular changes that occur during the movement of myosin cross-bridges.

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Recognition of Foreign Molecules by the Immune System

15

The immune system in vertebrates provides a defense mechanism against foreign parasites such as viruses and bacteria. Three main properties are essential to its successful operation: specific recognition of foreign molecules, the ability to destroy the foreign parasite, and a memory mechanism that allows a more rapid response to a second infection by the same microorganism.

Foreign invaders are recognized through specific and tight binding of the proteins of the immune system to molecules specific to the foreign organisms. The sites on foreign molecules that are recognized by the immune system are called **antigenic determinants**, and they interact with two different classes of antigen receptors produced by two major cell types of the immune system. **Antibodies**, also known as **immunoglobulins**, are produced by B cells, which are stimulated by antigen binding to secrete antibodies into the bloodstream (Figure 15.1a). Related molecules known as **T-cell receptors**, found on the surface of T cells, recognize and destroy virus-infected cells and play an important role in coordinating the immune response (Figure 15.1b).

Antibodies secreted by B cells bind to foreign material (antigen) and serve as tags or identifiers for such material. Antibody-tagged bacteria,

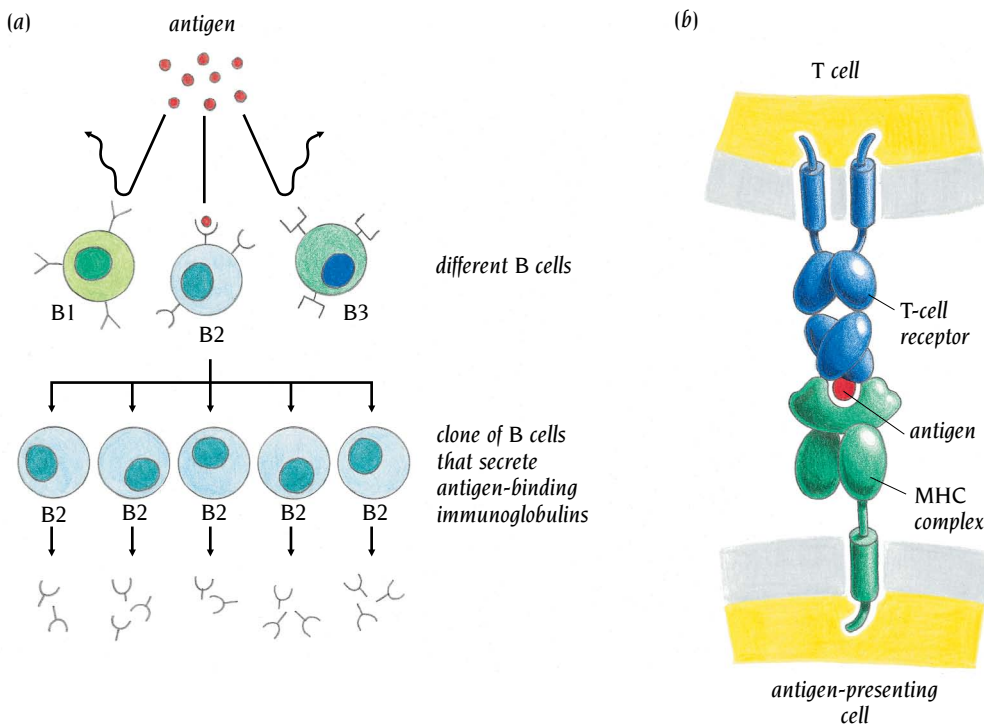


Figure 15.1 (a) The clonal selection theory. An antigen (red) activates those B cells that have immunoglobulin molecules on their surfaces that can recognize and bind the antigen. This binding triggers production of a clone of identical B cells that secrete soluble antigen-binding immunoglobulins into the bloodstream. (b) T cells recognize foreign viral antigens (red), through a T-cell receptor (blue) that can bind degraded fragments of the antigen when they are associated with an MHC molecule (green). The polypeptide chains of both T-cell receptors and MHC molecules are folded into immunoglobulin-like domains, represented as ellipsoids in the diagram.

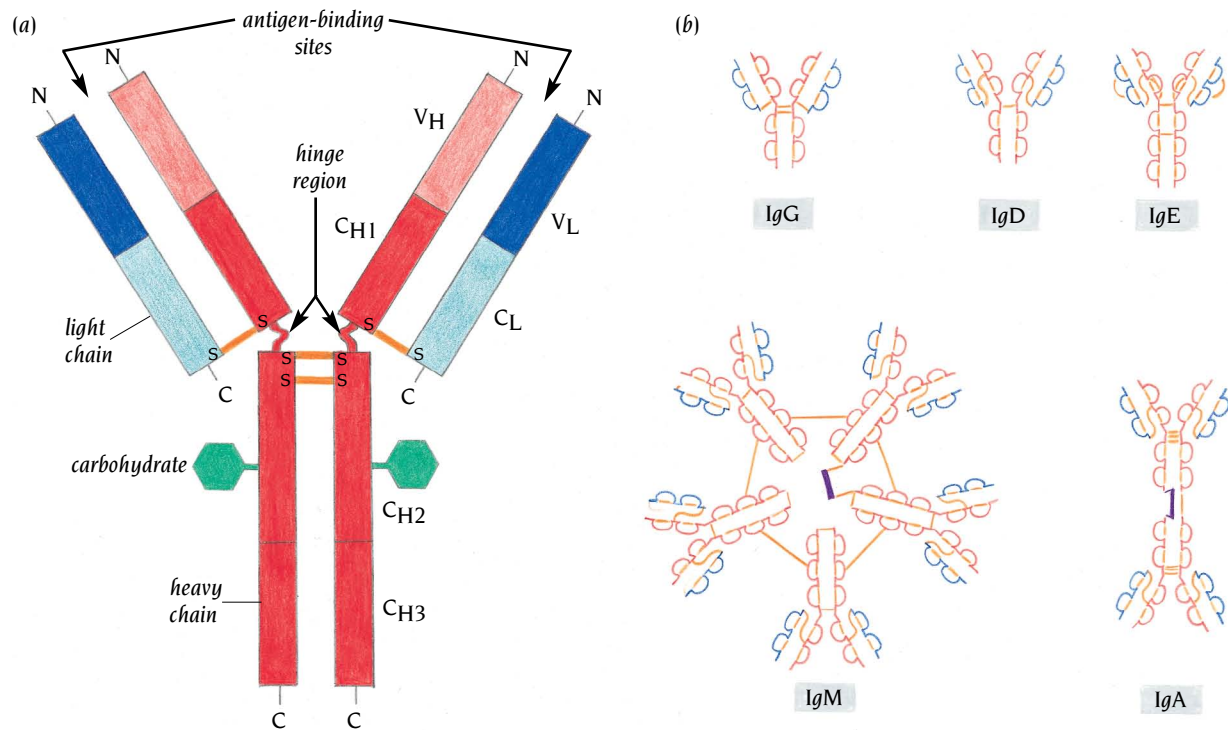
pathogens, or other foreign objects are recognized and disposed of by macrophages and other effector cells in the immune system; in addition, antibody-tagged bacteria can be directly lysed by a set of nonspecific serum proteins collectively called complement (because they are required in addition to antibody in triggering cell lysis). B cells rearrange the genes that code for their antibody proteins, so that each cell makes a unique antibody. B cells can display on their surface a membrane-bound form of their antibody. Binding of antigen to this form triggers a process that eventually activates the B cell to synthesize and secrete many copies of the antibody. In this way a small amount of antigen can elicit an amplified and specific immune response that helps to clear the body of the source of the antigen.

By contrast, T-cell receptors recognize antigenic determinants only when presented as part of a complex with a third important group of proteins, the **MHC molecules**, which fall into two classes, I and II. These proteins are encoded by the *major histocompatibility gene complex*, a region of chromosome six (in humans) originally identified as the genetic element that controls transplant rejection. Each individual carries a discrete set of MHC molecules, and differences in this set from one individual to another cause a transplanted organ to appear foreign to the immune system. MHC molecules bind degraded fragments of peptide antigens, generated inside infected cells or taken up by specialized phagocytotic cells, and display the fragments at the cell surface for recognition by T cells. T cells also rearrange their receptor genes so that the receptor carried by each T cell is unique. Binding of a T-cell receptor to an MHC–antigen complex triggers activation of the T cell to kill the infected cell, or elicits help from other cells in generating an immune response against the phagocytosed material.

In this chapter we will see that all three of the molecules directly involved in the specific recognition of antigen—immunoglobulins, the T-cell receptor, and the molecules of the MHC—belong to a family of proteins that seems to have evolved by duplication and diversification from a single ancestral domain. Because antibodies are naturally produced in very large quantities as soluble molecules, the crystal structure of an immunoglobulin was solved long before the structure of either of the other two molecules could be experimentally approached: hence the presumed ancestral domain structure is known as an immunoglobulin or immunoglobulin-like domain. MHC proteins and T-cell receptors are cell surface proteins that occur naturally in low abundance. Since the development of recombinant sources of protein for crystallization, structures have been determined for soluble forms of MHC proteins and T-cell receptors. We shall examine in detail the three-dimensional structures of antibodies and their bound antigens, and discuss the relationship between the genetic basis for antibody diversity and the structural basis for specific recognition. We shall also examine the structure of an MHC molecule and show that the structural principles for recognition and binding of antigens to MHC are quite different from those of antibodies and that they reflect the special requirement for presentation of bound antigen to the T-cell receptor. We will then look at the structure of a T-cell receptor and its special adaptations for binding to its ligand, which is the complex of an MHC molecule with bound antigen. Finally, we shall discuss the use of the immunoglobulin-like domain in other proteins of immunological interest.

The polypeptide chains of antibodies are divided into domains

The basic structure of all immunoglobulin (Ig) molecules comprises two identical **light chains** and two identical **heavy chains** linked together by disulfide bonds (Figure 15.2a). There are two different classes, or isotypes, of light chains, λ and κ , but there is no known functional distinction between them. Heavy chains, by contrast, have five different isotypes that divide the immunoglobulins into different functional classes: IgG, IgM, IgA, IgD, and IgE, each with different effector properties in the elimination of antigen



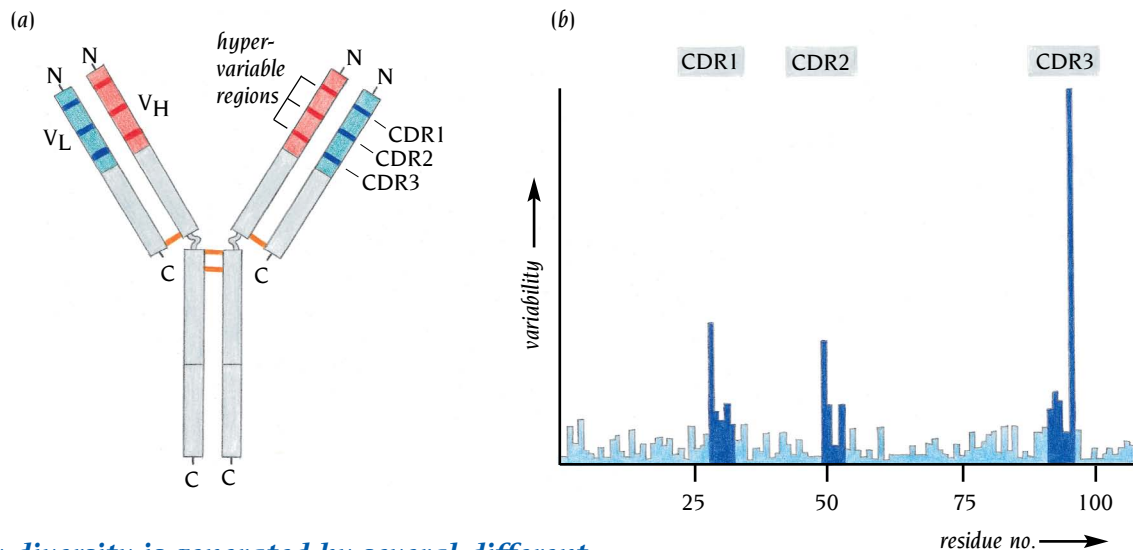
(Figure 15.2b). Each class of heavy chains can combine with either of the two different classes of light chains.

In this chapter we will discuss immunoglobulins of the **IgG** class, which is the major type of immunoglobulin in normal human serum, and which has the simplest structure. Each chain of an IgG molecule is divided into domains of about 110 amino acid residues. The light chains have two such domains, and the heavy chains have four.

The most remarkable feature of the antibody molecule is revealed by comparing the amino acid sequences from many different immunoglobulin IgG molecules. This comparison shows that between different IgGs the amino-terminal domain of each polypeptide chain is highly variable, whereas the remaining domains have constant sequences. A light chain is thus built up from one amino-terminal **variable domain** (V_L) and one carboxy-terminal **constant domain** (C_L), and a heavy chain from one amino-terminal variable domain (V_H), followed by three constant domains (C_{H1}, C_{H2}, and C_{H3}).

The variable domains are not uniformly variable throughout their lengths; in particular, three small regions show much more variability than the rest of the domain: they are called **hypervariable regions** or **complementarity determining regions**, CDR1–CDR3 (Figure 15.3). They vary both in size and in sequence among different immunoglobulins. These are the regions that determine the specificity of the antigen–antibody interactions. The remaining parts of the variable domains have quite similar amino acid sequences. In fact, all variable domains show significant overall sequence homology and, hence, evolutionary relationships among each other. Similarly, the different constant domains—C_L, C_{H1}, C_{H2}, and C_{H3}—show significant (30–40%) sequence identity with each other and with constant domains from immunoglobulin chains of different classes. This strongly suggests that the genes that code for these constant domains arose by successive gene duplication of an ancestral antibody gene. In contrast, there is no significant sequence homology between constant and variable domains, even though, as we will see, there are striking structural similarities. Before describing the structures of the immunoglobulin domains, however, we pause to outline the genetic mechanisms underlying the variability of immunoglobulin molecules.

Figure 15.2 (a) The immunoglobulin molecule, IgG, is built up from two copies each of two different polypeptide chains, heavy (H) and light (L). The L chain folds into two domains: V_L with variable sequence between different IgG molecules and C_L with constant sequence. The H chain folds into four domains: one variable V_H and three constant domains, C_{H1}, C_{H2}, and C_{H3}. Disulfide bridges connect the four chains. The antigen-binding sites are at the ends of the variable domains. (b) Schematic polypeptide chain structure of different immunoglobulin molecules. IgM is a pentamer where the heavy chains have one variable domain and four constant domains. An additional polypeptide chain, the J chain, (black bar) is associated with the pentameric molecule. The J chain also links two units to form the dimeric IgA. IgG, IgD, and IgE are monomeric immunoglobulin molecules.



Antibody diversity is generated by several different mechanisms

The human body generates as many as 50 million new antibody-producing B cells every day. The vast majority of these cells produce an antibody molecule unique to that cell and its progeny. Genetic information for this diversity is contained in more than 1000 small segments of DNA. One of the outstanding achievements of modern biology has been the elucidation of the mechanisms that generate antibody diversity.

The gene segments encoding the variable and constant regions of antibodies are clustered in three gene pools, one for the heavy chain and one for each of the two light-chain isotypes, on separate chromosomes. In the heavy-chain gene pool the variable domain is encoded by three types of segments, V, D, and J (Figure 15.4). V codes for approximately the first 90 residues, D for the hypervariable region CDR3, and J for the remaining 15 residues of the variable domain.

There are about 1000 different V segments, about 10 different D segments of variable lengths, and about 4 different J segments in the heavy-chain gene pool. The DNA for the variable domain of a new B cell is assembled by random joining of one of each of these segments into a single continuous exon (Figure 15.5), a process called combinatorial joining that can occur in about 40,000 different ways. The joining of these segments is not precise at the junction of D with V and J, and this creates additional junction diversity. Extra nucleotides can be added during this joining procedure, giving still further diversity. The D segment and nearby junctional regions encodes CDR3, which is thus a focus of particular diversity in the heavy chain.

The newly created V-D-J exon becomes joined to one of the eight C segments that encode the constant regions of the various classes of heavy chain. Usually, a variable region is first expressed in an IgM molecule. Later, further genetic rearrangements will delete the mu (μ) exons encoding the IgM constant region, and the variable region will become joined to a second set of constant-region exons encoding IgA, IgD, IgE, or most commonly IgG. This exchange of variable regions among heavy chains is called **class-switching**.

The genetic mechanisms underlying light chain diversity are similar except that there are no D segments, and the diversity of CDR3 therefore depends largely upon junctional diversity generated during V-J joining.

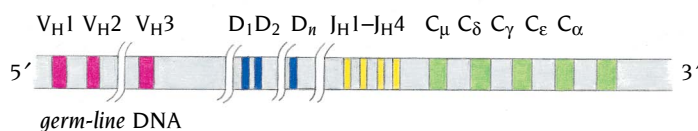


Figure 15.3 Certain regions within the 110 amino acid variable domains show a high degree of sequence variation. These regions determine the antigen specificity and are called hypervariable regions or complementarity determining regions, CDR (a). There are three such regions, CDR1-CDR3, in each variable domain. The sequences of a large number of variable domains in L chains have been compared, and the sequence variability is plotted as a function of residue number along the polypeptide chain in (b). The three large peaks in this diagram correspond to the three hypervariable regions in (a): CDR1-CDR3. [(b) Adapted from T.T. Wu and E.A. Kabat, *J. Exp. Med.* 132: 211-250, 1970.]

Figure 15.4 Variable domains in immunoglobulins (and T-cell receptors) are made by combinatorial joining of gene segments. Three such segments, V, D, and J, are joined to make the variable domain of a heavy chain. In the mouse the gene-segment pool for an H chain contains about 1000 V segments, 12 D segments, 4 J segments, and a cluster of C segments, each encoding a different class of H chain.

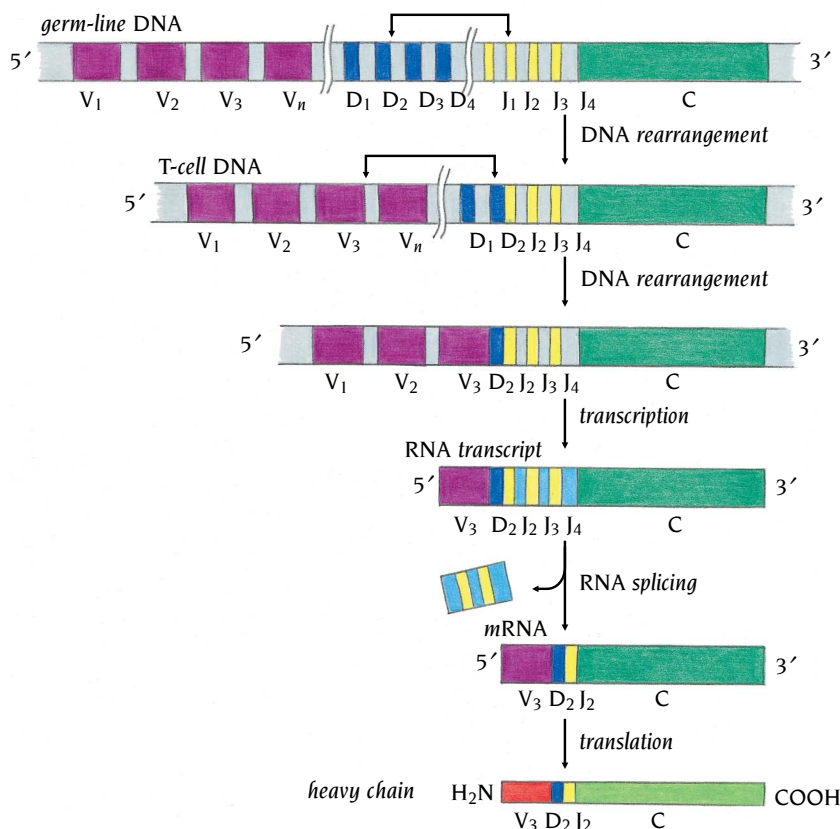


Figure 15.5 The V-D-J joining process involved in making a heavy chain in the mouse. In the germ-line DNA the cluster of about 4 J gene segments and about 10 D segments are separated from the C gene segment by a short intron and from the about 1000 V gene segments by long regions of DNA. During B cell development the chosen V gene segment (V_3) is moved to lie precisely next to the chosen D and J segments (D_2 and J_2). The “extra” J genes (J_3 and J_4) and the intron sequence are transcribed along with the V_3 , D_2 , J_2 , and C segments and then removed by RNA splicing to generate mRNA molecules.

By these genetic mechanisms we can estimate that at least 90,000 different heavy chains and 3000 different light chains can be produced. We will see that it is reasonable to assume from structural considerations that almost any heavy chain can combine with almost any light chain to produce a functional antibody. Since the antigen-binding sites are built up from the CDR regions of both chains, this ability of light and heavy chains to combine and form functionally viable pairs generates considerable additional diversity. By this combinatorial association, the number of different antibody molecules a new B cell can choose from is increased to $3000 \times 90,000 = 270$ million. The actual number of antibody molecules with different antigen binding sites is, however, several orders of magnitude higher because of yet one more mechanism for increasing diversity in mature B cells. In these cells, point mutations can be introduced into the exons for the variable domains by somatic hypermutation.

All immunoglobulin domains have similar three-dimensional structures

Complete IgG molecules have been difficult to crystallize because of the conformational flexibility in the **hinge region** between C_{H1} and C_{H2} of the heavy chains (see Figure 15.2). When an IgG molecule is partially digested with the proteolytic enzyme papain, however, the heavy chains are cleaved in the hinge region to give two identical Fab fragments (for *F*ragment, *a*ntigen *b*inding) and one Fc fragment (for *F*ragment, *c*rystallizes easily) (Figure 15.6). The Fab molecule comprises one complete light chain linked by a disulfide bridge to a fragment of the heavy chain consisting of V_H and C_{H1} . The Fc molecule comprises C_{H2} and C_{H3} from both heavy chains, which are also linked by disulfide bridges. Until recently, all detailed high-resolution x-ray structural information on immunoglobulins has been obtained from structure determinations of Fab, Fc, or similar fragments including those of complexes with antigen.

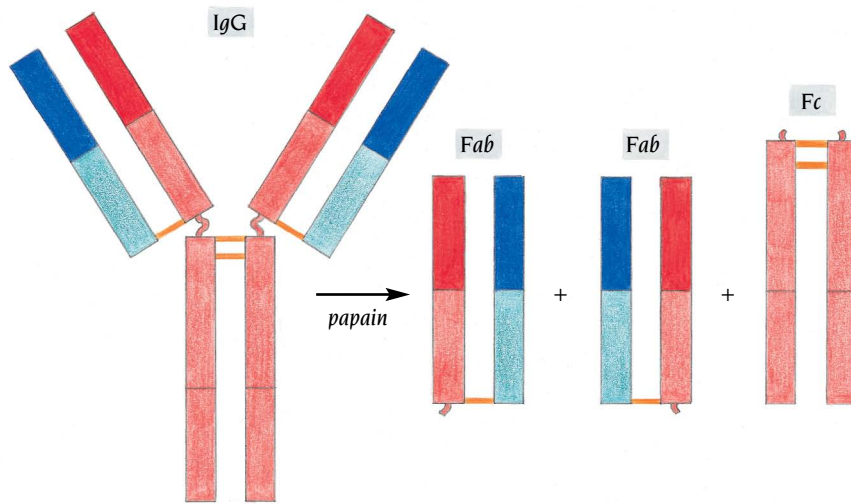


Figure 15.6 Enzymatic cleavage of immunoglobulin IgG. The enzyme papain splits the molecule in the hinge region, yielding two Fab fragments and one Fc fragment.

Before the advent of monoclonal antibody technology, Allen Edmundson and collaborators at Argonne National Laboratories, US, obtained the first detailed description of an immunoglobulin domain. The high-resolution structure they determined in 1973 was that of a Bence-Jones protein, a tumor-derived molecule that is a dimer of two light chains. In the same year Roberto Poljak and collaborators at Johns Hopkins Medical School, Baltimore, determined the first high-resolution structure of an intact immunoglobulin Fab fragment, and David Davies and collaborators at the National Institutes of Health demonstrated the binding of an antigenic determinant to the antigen-binding site in a Fab fragment. Since then the structures of over one hundred various other proteolytic fragments of immunoglobulin molecules, alone or in complex with antigen, have been solved to high resolution. In all cases both constant and variable domains from both heavy and light chains have proved to have similar structures, which has become known as the **immunoglobulin fold**.

The immunoglobulin fold is best described as two antiparallel β sheets packed tightly against each other

All constant domains are built up from seven β strands arranged so that four strands form one β sheet and three strands form a second sheet (Figure 15.7). The sheets are closely packed against each other and joined together by a disulfide bridge from β strand F in the three-stranded sheet, to β strand B in the four-stranded sheet. The topology of the sheets is quite simple, as can be seen from the topological diagram in Figure 15.7b. Even though the actual structure consists of two separate β sheets, it is convenient to regard the topology as a form of a Greek key barrel.

The loops are short, and as a result the majority of the residues of the domain are in the two β sheets. These framework residues have the same structure in every constant domain studied, and the structure is the same in both heavy and light chains. Most of the invariant residues of the constant domain, including the disulfide bridge, are found in the sheets of the framework region. These invariant residues have two important functions. One is to form and stabilize the framework by packing the β sheets through hydrophobic interactions to give a hydrophobic core between the sheets; the other is to provide interactions between constant domains of different chains to form a complete immunoglobulin molecule.

The residues not in the framework region form the loops between the β strands. These loops may vary in length and sequence among immunoglobulin chains of different classes but are constant within each class; the sequence of the loops is invariant. The functions of these loops are not known, but they are probably involved in the effector functions of antibodies. When an antibody-antigen complex has been formed, signals are

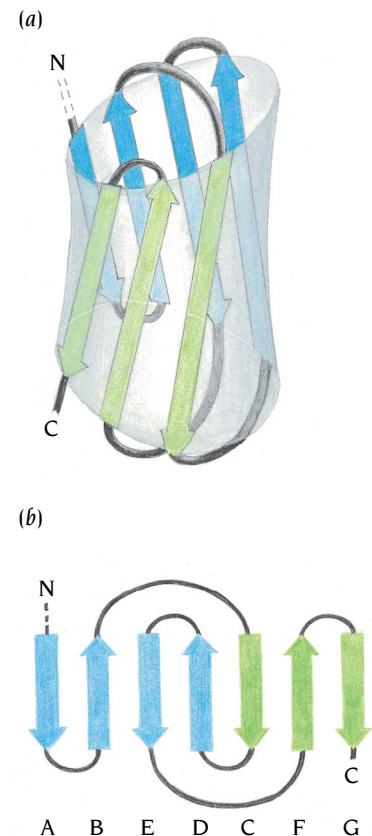


Figure 15.7 The constant domains of immunoglobulins are folded into a compressed antiparallel β barrel built up from one three-stranded β sheet packed against a four-stranded sheet (a). A topological diagram (b) shows the connected Greek key motifs of this fold.

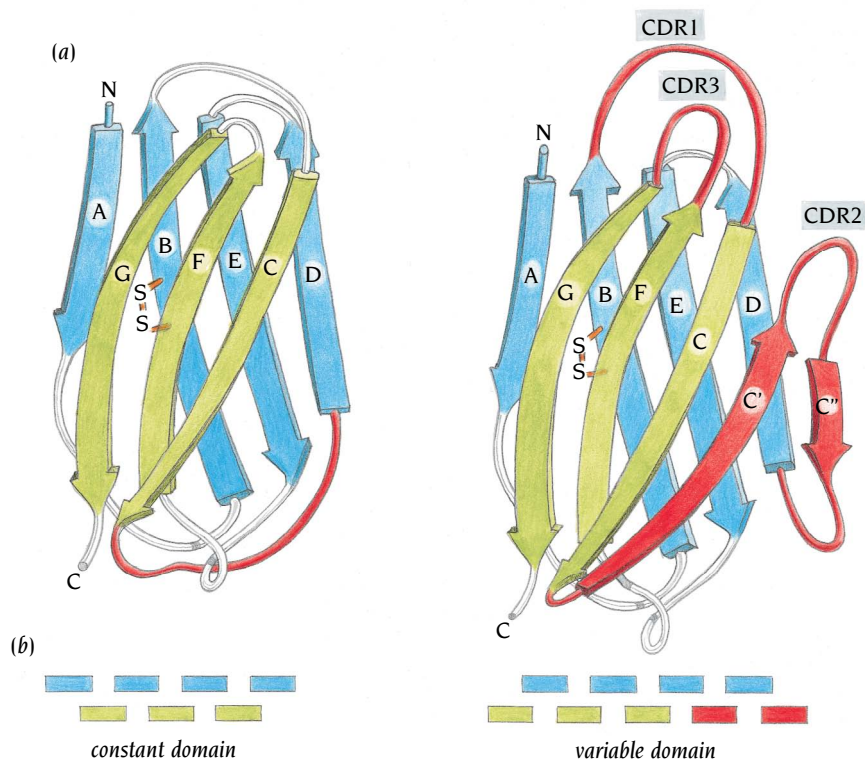


Figure 15.8 (a) Comparison of the structures of the constant and variable domains of immunoglobulins. Beta strands labeled A–G have the same topology and similar structures. There are two extra β strands, C' and C'' (red) in the variable domain. The loop between these strands contains the hypervariable region CDR2. The remaining CDR regions are at the same end of the barrel in the loops connecting β strands B and C and strands F and G. A disulfide bond bridges strand B in one sheet with strand F in the other sheet in both the constant and the variable domains. (b) The β strands viewed end on, illustrating that one β sheet has the same four β strands in the two domains.

mediated through the Fc region to different systems, such as phagocytic cells or the complement system, which will destroy the complex. These activities are triggered by specific interactions of ligands with the constant domains of immunoglobulins.

The overall structure of the variable domain is very similar to that of the constant domain, but there are nine β strands instead of seven. The two additional β strands are inserted into the loop region that connects β strands C and D (red in Figure 15.8). Functionally, this part of the polypeptide chain is important since it contains the hypervariable region CDR2. The two extra β strands, called C' and C'', provide the framework that positions CDR2 close to the other two hypervariable regions in the domain structure (Figure 15.8).

The hypervariable regions are clustered in loop regions at one end of the variable domain

The specificity of immunoglobulins is determined by the sequence and size of the hypervariable regions in the variable domains. From amino acid sequence comparisons of several hundred different variable domains, these hypervariable regions have been defined as amino acids 24–34, 50–56, and 89–97 in the light chains and 31–35, 50–65, and 95–102 in the heavy chains. In the three-dimensional structures these residues occur in the loop regions that connect β strands B–C, C'–C'', and F–G. We can see in Figure 15.8 that these loop regions are clustered together at one end of the β sheets. CDR2 and CDR3 are hairpin loops between β strands C'–C'' and F–G in the five-stranded β sheet. The third hypervariable region CDR1 is a cross-over connection from β strand B in the four-stranded sheet to strand C in the five-stranded sheet. All strands in the five-stranded sheet thus contribute loops to the hypervariable regions, whereas only one strand from the four-stranded sheet is involved. The additional loop at this end of the sheet, between β strands D and E in the four-stranded sheet, does not have a variable sequence and does not participate in antigen binding. It is thus evident that loops from the five-stranded β sheet contribute more residues to the complete antigen-binding site than the four-stranded sheet.

The variable domains of immunoglobulins are excellent examples of the important structural principle discussed in Chapter 4 in connection with α/β -barrel structures. Functional residues in protein molecules are frequently provided by loop regions attached to one end of a stable structural framework that is built up mainly by secondary structure elements. In α/β -barrel structures this framework is built from eight parallel β strands surrounded by eight helices, whereas in the variable domains of immunoglobulins it consists of nine antiparallel β strands arranged in two sheets packed against each other. The fact that there exists a great number of sequence variations in the loop regions of antibodies demonstrates that the folding of the structural framework is quite insensitive to changes in the functional loop regions. Thus, it has been possible to construct, using recombinant DNA methods, chimeric antibodies carrying CDR loop regions from one antibody, and framework regions from another antibody.

The antigen-binding site is formed by close association of the hypervariable regions from both heavy and light chains

Fab fragments contain one arm of the IgG molecule with an intact antigen binding site. In these molecules, as in intact IgG, the light chain is associated with the heavy chain in such a way that C_L associates with C_{H1} and V_L with V_H (Figure 15.9). These associations are very tight and extensive. The segments of the light and heavy polypeptide chains that join V_L to C_L and V_H to C_{H1} have few contacts. As a result, the Fab fragment consists of two globular regions, one formed by the two constant domains and the other by the two variable domains. Together, they make an elongated molecule. The hypervariable regions of both the light and heavy chains are close together

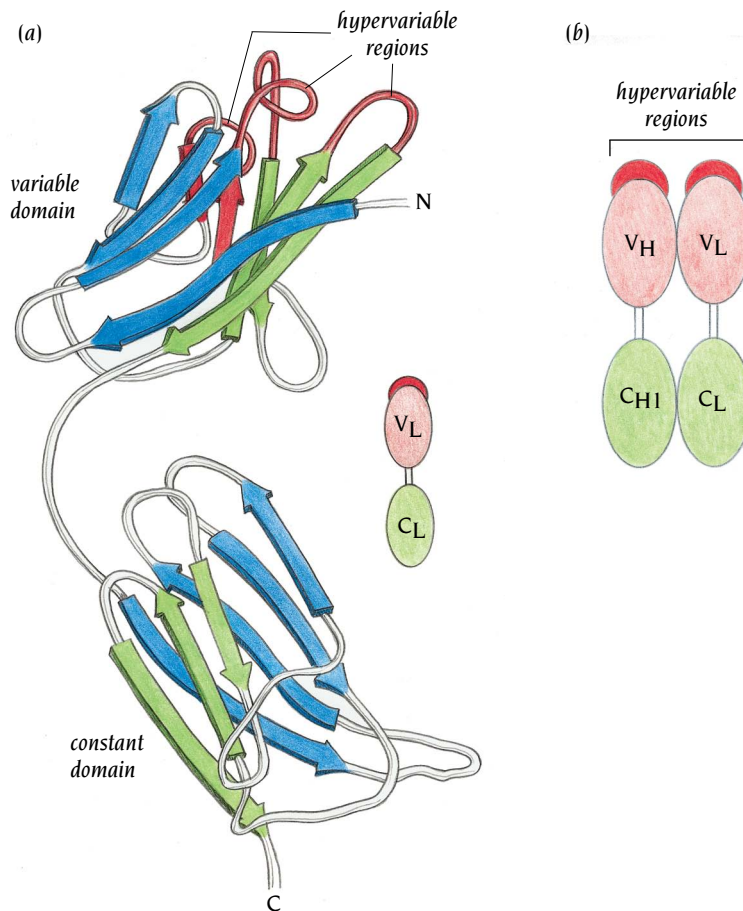


Figure 15.9 (a) The variable and constant domains in the light chain of immunoglobulins are folded into two separate globular units. In both domains the four-stranded β sheet is blue and the other sheet is yellow. The hypervariable CDR regions are at one end of this elongated molecule. (b) In the Fab fragment, as well as in the intact immunoglobulin molecule, the domains associate pairwise so that V_H interacts with V_L and C_{H1} with C_L . By this interaction the CDR regions of both variable domains are brought close to each other and together form the antigen-binding site.

Figure 15.10 Schematic diagrams of the packing of the four-stranded β sheets of the constant domains C_{H1} and C_L in an Fab fragment of IgG. The sheets are viewed perpendicular to the β strands in (a) and end-on in (b), where the four-stranded β sheets are blue.

at one end of this elongated molecule. We will now examine more closely how these domains associate.

The constant domains associate by interactions between the four-stranded β sheets in both C_L and C_{H1} , which are closely packed with mainly hydrophobic residues in the interface. The directions of the β strands in the two domains are almost at right angles to each other (Figure 15.10a). This mode of packing β sheets against each other occurs frequently within the core of single domains that are built up from two β sheets—for example, in retinol-binding protein discussed in Chapter 5—but they are rather unusual for domain–domain associations. The effect of this packing is to bury the central regions of the four-stranded β sheets away from the solvent. Each four-stranded β sheet has the three-stranded β sheet of the same domain on one side and the four-stranded β sheet from the second domain on its other side (Figure 15.10b). This places rather stringent constraints on the sequence of these regions and most of the residues that occur in the interface are, in fact, conserved. In addition, residues that form the hydrophobic core within each domain are generally conserved.

The variable domains associate in a strikingly different manner. It is obvious from Figure 15.11 that if they were associated in the same way as the constant domains, via the four-stranded β sheets, the CDR loops, which are linked mainly to the five-stranded β sheet, would be too far apart on the outside of each domain to contribute jointly to the antigen-binding site. Thus in the variable domains the five-stranded β sheets form the domain–domain interaction area (Figure 15.11). Furthermore, the relative orientation of the β strands in the two domains is closer to parallel than in the constant domains and the curvature of the five-stranded β sheets is such that they do not pack

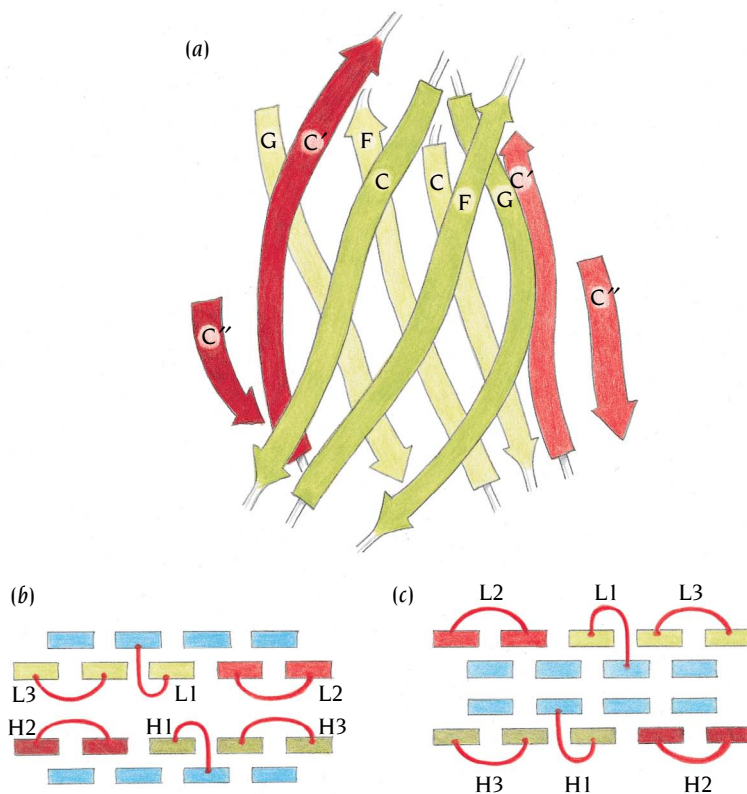
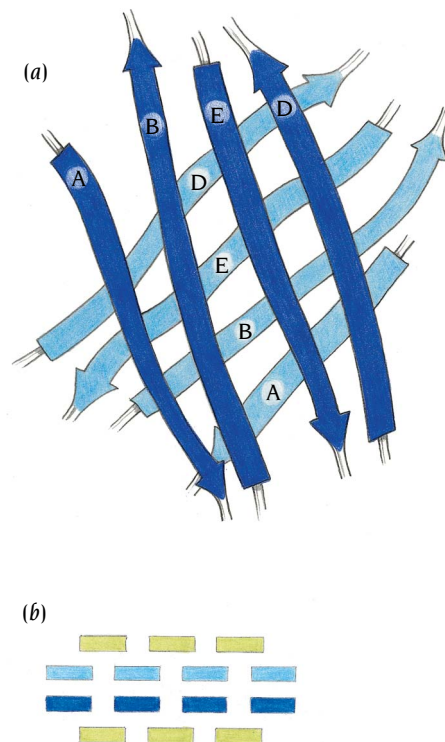


Figure 15.11 The two variable domains V_H and V_L are packed against each other so that the six hypervariable regions are close to each other. (a) Schematic diagram of the packing of the five-stranded β sheets of V_H and V_L in IgG. Only four of the β strands are involved in packing the variable domains against each other. Strand C'' is not involved. (b) Schematic diagram of the strands viewed end-on in the variable domain. Hypervariable loop regions are red. (c) Diagram illustrating a hypothetical packing of the variable domains through their four-stranded sheets. The hypervariable regions would be far apart.

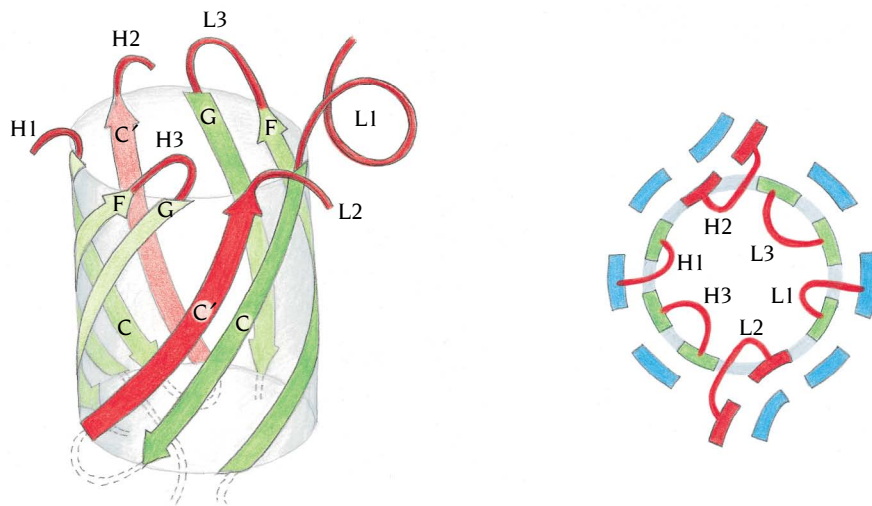


Figure 15.12 Schematic diagram of the barrel arrangement of four β strands from each of the variable domains in Fab. The six hypervariable regions, CDR1–CDR3 from the light chain (L1–L3) and from the heavy chain (H1–H3), are at one end of this barrel. (From J. Novotny et al., *J. Biol. Chem.* 258: 14433–14437, 1983.)

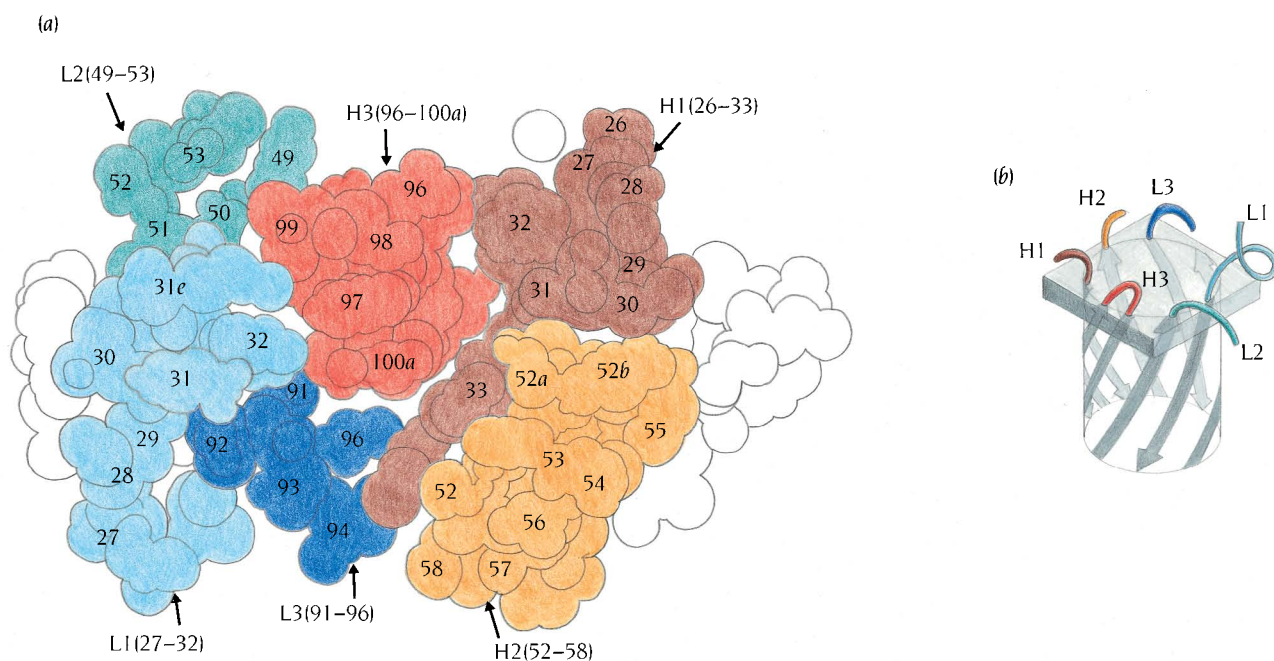
tightly against each other. Instead, each sheet forms half a barrel. When the two sheets associate, four of the five strands from both domains complete a barrel structure of eight antiparallel β strands with the six CDR loops at the same end of the barrel (Figure 15.12). Hydrophobic side chains fill the inside of this barrel.

The antigen-binding site binds haptens in crevices and protein antigens on large flat surfaces

The six CDR loops from the two variable chains at the rim of the eight-stranded barrel provide an ideal arrangement for generating antigen-binding sites of different shapes depending on the size and sequence of the loops (Figure 15.13). They can create flat extended binding surfaces for protein antigens or specific deep binding cavities for small hapten molecules.

Haptens, a special class of antigen, are small molecules that induce specific antibody production when they are attached to a protein that acts as a carrier. Phosphorylcholine is one such hapten that has been widely used in the investigation of immune responses. The specific binding of this hapten

Figure 15.13 (a) Drawing of a space-filling model of the hypervariable regions of an Fab fragment. The superpositions of five sections are shown, cut through a model as shown in (b). It is clearly seen that all six hypervariable regions (L1–L3, H1–H3) contribute to the surface shown here. (From C. Chothia and A. Lesk, *J. Mol. Biol.* 196: 901–917, 1987.)



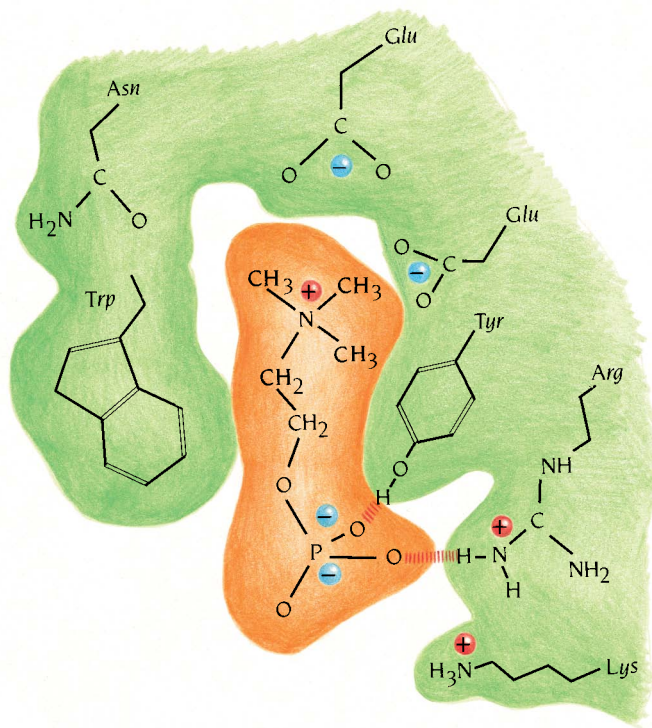


Figure 15.14 Schematic representation of the specific interactions between phosphorylcholine (orange) and the protein side groups (green) in Fab. The binding cavity is in a cleft between the light and the heavy chains. Choline binds in the interior while the phosphate group is toward the surface. (Adapted from E.A. Padlan et al., *Immunochemistry* 13: 945–949, 1976.)

to an Fab fragment has been studied crystallographically by David Davies and coworkers at National Institutes of Health. The binding cavity is about 15 Å wide at the mouth and 12 Å deep. It is lined by residues from all CDR loops except the short CDR2 from the light chain. A number of side chains from these loops bind phosphorylcholine in a manner that is very similar to that of a substrate or inhibitor binding to an enzyme (Figure 15.14).

In enzyme-catalyzed reactions, substrates in their transition state bind very tightly to the enzyme and are positioned close to polar or charged side chains that participate in catalyzing the enzymatic reaction. It has recently been possible to induce production of antibodies with enzymatic activity by choosing haptens that are analogs of the transition state of the substrate for a chemical reaction. When an animal is immunized with a transition-state analog as a hapten, an array of different antibodies that bind the hapten specifically are produced. Some of these antibodies happen to have charged or polar side chains suitably placed to promote catalytic reactions involving the corresponding substrate. These catalytic antibodies can be isolated and improved upon by recombinant DNA techniques.

The analysis of hapten binding to antibodies was possible before monoclonal antibodies became available because some myeloma proteins bind haptens. Thus hapten binding was studied as early as 1974. Once monoclonal antibodies with any chosen specificity could be produced, it became possible in principle to study the interactions of antibodies with protein antigens. However, the path from a cell line to a high-resolution three-dimensional antigen-antibody structure is long, and not until 1986 was the first such investigation reported by Roberto Poljak and coworkers at the Pasteur Institute, France. They chose a lysozyme–Fab complex because the structure of the enzyme lysozyme was known and its antigenic properties had been extensively studied.

The shape of the interaction area between lysozyme and the CDR loops of the antibody is easily distinguished from the hapten-binding crevice. The interaction extends over a large area with maximum dimensions of about 20 × 30 Å (Figure 15.15). The interaction surface is irregular but relatively flat, with small protuberances and depressions that are complementary in the antigen and the antibody. Residues from all six CDR loops contribute to the

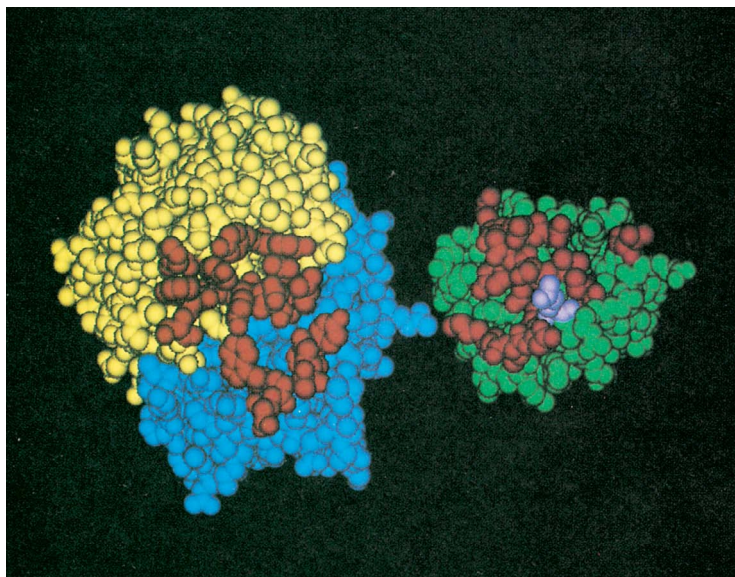


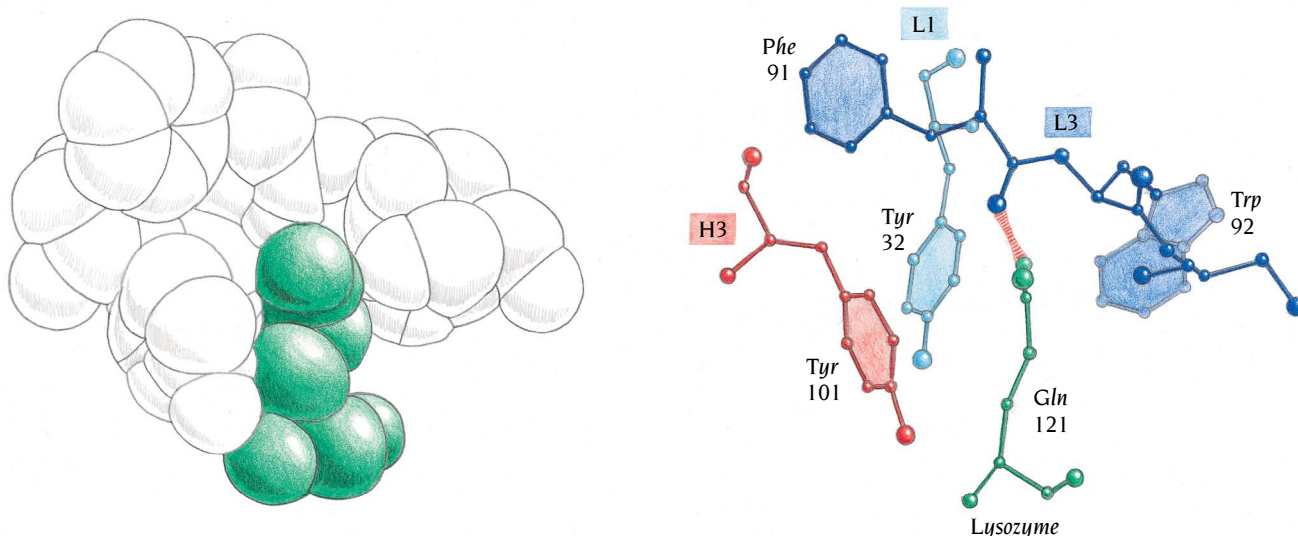
Figure 15.15 Space-filling representation of a complex between lysozyme (green) and the Fab fragment of a monoclonal antilysozyme (blue and yellow). The Fab fragment and the antigen (lysozyme) have been separated in this diagram, and their combining surfaces are viewed end-on. Atoms that are in contact in the complex are colored red both in Fab and lysozyme, except Gln 121 in lysozyme, which is violet. The diagram illustrates the large size of the interaction surfaces. (After A.G. Amit et al., *Science* 233: 747–753, 1986; courtesy of R. Poljak.)

antibody surface, and residues from two stretches of the polypeptide chain of lysozyme, 18–27 and 116–129, form the antigen surface. Seventeen antibody residues make close contact with 16 lysozyme residues. These residues pack against each other with a density similar to that found in the interior of protein molecules. For example, Gln 121 in lysozyme (violet in Figure 15.15) protrudes from the lysozyme surface and fits snugly into a cleft between the CDR3 loops of V_H and V_L (Figure 15.16). On the other hand, there are some imperfections, in the form of holes, in the interaction area. Appropriate changes in the CDR loops could fill these holes and produce an even better fit. Presumably, such adjustment could occur through somatic mutations in the antibody genes to give rise to higher-affinity antibodies.

The interaction area is centered on CDR3 of the heavy chain, which makes a proportionally greater contribution to the formation of this area than any of the other CDR loops. Four residues in the heavy-chain CDR3 participate in interactions with lysozyme, and they are all encoded in the D segment, the special significance of which now becomes clear. CDR3 of the heavy chain is the principal focus of the somatic mechanisms for generating antibody diversity and is central to the specific recognition of protein antigens. It is also the loop that eludes the structural predictive formula of Chothia and Lesk that we describe in the following section.

The structure of lysozyme in the complex is the same as that in crystals of free lysozyme, and no conformational changes are seen even in the regions

Figure 15.16 Detailed views of the environment of Gln 121 in the lysozyme–antilysozyme complex. Gln 121 in lysozyme is colored green both in the space-filling representation to the left and in the ball and stick model to the right. This side chain of the antigen fits into a hole between CDR3 regions of both the heavy (Tyr 101) and the light (Trp 92) chains as well as CDR1 from the light chain (Tyr 32). (After A.G. Amit et al., *Science* 233: 747–753, 1986.)



that bind to the antibody. The structure of the Fab fragment complexed to lysozyme is also similar to its structure when not bound to antigen. However, in other cases, conformational changes that increase the complementarity between the interacting surfaces of antigen and antibody can accompany antigen binding. Adjustments ranging from simple side chain rotations, to CDR loop reorientations of up to 6 Å, to rigid body domain motions of up to 15°, have been observed for various complexes. Although antibody–antigen interactions have been always at the forefront of “lock-and-key” versus “induced-fit” debate about the physical origin of biological specificity, no general rules have yet emerged.

The CDR loops assume only a limited range of conformations, except for the heavy chain CDR3

We have discussed how antibodies are able to bind a wide range of antigens and even catalyze chemical reactions through variations in their hypervariable CDR loops, which are attached to the essentially invariant β -sheet framework of the immunoglobulin domain. Therefore, in order to engineer an antibody with a predetermined binding specificity, or to predict the binding specificity of an antibody with known sequence but unknown function, we must be able to predict and model accurately the conformations of CDR loops. Thus the question arises: can the structure of CDR loops vary randomly, or are there certain preferred conformations that can be deduced from the lengths and sequences of these regions?

Cyrus Chothia and Arthur Lesk at MRC, Cambridge, UK, have attempted to answer this question by an analysis of the CDR-loop conformations in known structures. For each observed conformation they identified those residues that were mainly responsible for maintaining that conformation, either through interactions with framework residues or through their ability to assume unusual conformations. Examples of the latter are Gly or Pro. Examination of the sequences of variable domains of unknown structure showed that many have CDR loops that are similar in size to those of one of the known structures, with identical residues at the sites responsible for the observed conformation. Thus, for these loops, conformations could reliably be predicted. In addition to their value in the modeling and design of new antibodies, these results have interesting implications for the molecular mechanisms involved in the generation of antibody diversity. For five of the six hypervariable regions of most immunoglobulins (CDR3 of the heavy chain is an exception) there appears to be only a small repertoire of main-chain conformations, most of which are known from the set of immunoglobulin structures so far determined. Sequence variations within the hypervariable regions modulate the surface that these canonical loop structures present to the antigens. Sequence variations within both the framework and the hypervariable regions shift the canonical structures relative to each other by small but significant amounts.

Three examples from the analysis by Chothia and Lesk, which is further described in Chapter 17, demonstrate these phenomena. The loop between β strands C' and C'' in CDR2 of the light chain is three residues long in all known sequences. This loop has very similar conformations in the known structures, and 95% of 250 known sequences of mouse and human light chains fit the sequence constraints of this structure. There is thus a high probability that this loop has the same conformation in almost all immunoglobulins. Four different conformations of CDR2 from the heavy chain have been found in the 10 x-ray structures that were analyzed. For each of these conformations there are specific constraints at a few residues. For 236 of 300 known sequences of heavy chains, the sequence of the CDR2 loop fits one of these constraints and would therefore be expected to have one of these four canonical CDR2-loop structures. The CDR3 loop of the heavy chain, however, varies in length from 6 to 14 residues in the known sequences and forms a long hairpin loop. For such large loops many conformations are possible. The one actually found will depend both on the size

and sequence of the loop and on its packing against the rest of the protein. This loop, therefore, is difficult to predict and model correctly.

An IgG molecule has several degrees of conformational flexibility

Early physicochemical and electron microscopy studies suggested that in the intact IgG molecule the Fab arms were flexible and could move relative to each other and to the Fc stem. The region of the heavy chains that connects C_{H1} with C_{H2} , (and therefore Fab with Fc) is called the hinge region. The amino acid sequence of this region includes the cysteine residues that form disulfide bridges between heavy chains, and has a high proportion of proline residues frequently found in flexible regions of proteins. The flexibility of the hinge region is important for the function of immunoglobulins, but it complicates crystallization, and in many cases crystallized intact antibodies have disordered Fc and hinge regions that are not visualized in the crystal structure.

The structure of an intact antibody, an antitumor IgG that forms unusually well-ordered crystals, was determined by Alexander McPherson and colleagues in 1992 (Figure 15.17). All segments of the antibody, including the hinge domain and attached carbohydrate, are visible in the electron density maps. As expected from earlier studies of Fc domains, the C_{H2} and C_{H3} domains of each of the heavy chains associate in C_{H2} – C_{H2} and C_{H3} – C_{H3} pairs. The C_{H3} pair is associated in the same way as the C_{H1} – C_L pair discussed previously. In contrast, there is no protein–protein interaction between the C_{H2} domains: instead, a carbohydrate chain is attached to each C_{H2} domain in the interface region and forms a weak bridge between them. Polypeptide segments between the Fab and Fc portions are extended (compare Figures 15.2a and 15.17) and exhibit large temperature factors indicative of a high degree of mobility. Thus, the hinge “domain” is better considered as a flexible tether approximately 50 Å long than as a rigid hinge. There is no overall symmetry to the molecule: the local twofold rotation axis relating the Fc C_{H3} – C_{H3} domains is different from that relating the Fab C_{H1} – C_L domains. The orientation of the two Fab arms is such that the intact IgG molecule resembles a “T”. However, studies of other intact antibodies by crystallography and NMR spectroscopy indicate that there is no preferred orientation of the arms. Because the two Fab arms can move relative both to each other and to Fc, the two antigen-binding sites at the tips of the Fab arms can move to bind antigenic determinants separated by different distances and in different orientations, and the Fc region can move relative to the Fab to facilitate interaction with effector molecules that bind Fc.

Structures of MHC molecules have provided insights into the molecular mechanisms of T-cell activation

Few protein crystal structures have had such a direct impact in biology as that of the human class I major histocompatibility complex (MHC) molecule, **human lymphocyte antigen A2 (HLA-A2)**, determined in 1987 by Pamela Björkman and others in Don Wiley’s group at Harvard University. MHC molecules are crucial to the activation of T cells, which play a major role in immune defense. Many years of detailed molecular genetic analysis had established that T-cell activation is initiated when the antigen receptors on the surface of these cells come into contact with foreign peptide antigen bound to MHC molecules on the surface of antigen-presenting cells. Several major questions arose from these observations. How can a few different MHC molecules recognize and bind an almost unlimited number of antigens? How are these antigens presented to T-cell receptors so that they discriminate between self and nonself? Do the T cells see antigen and MHC molecules, which are firmly attached to the surface of the antigen-presenting cells, as a single structure or as two separate entities?

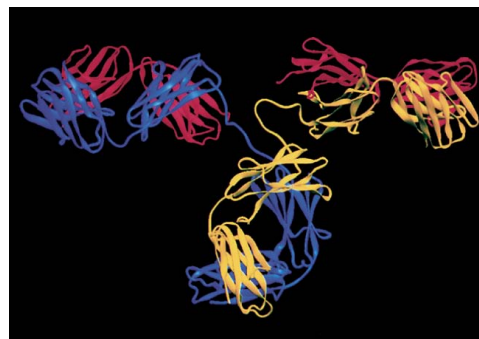


Figure 15.17 The three-dimensional structure of an intact IgG. Hinge regions connecting the Fab arms with the Fc stem are relatively flexible, despite the presence of disulfide bonds in this region linking the heavy and light chains. Carbohydrate residues that bridge the two C_{H2} domains are not shown. (Courtesy of A. McPherson and L. Harris, *Nature* 360: 369–372, 1992, by copyright permission of Macmillan Magazines Limited.)

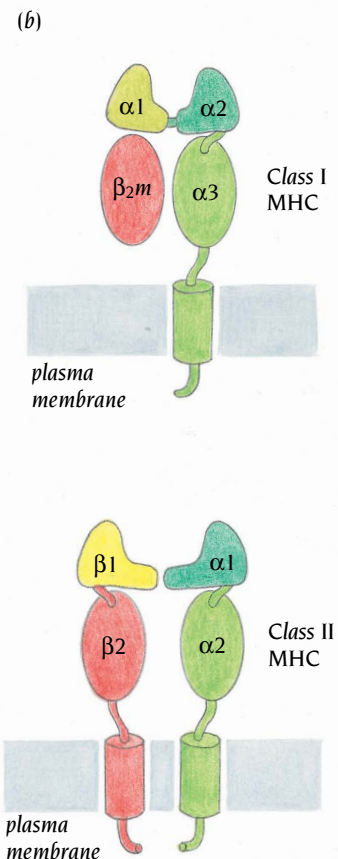
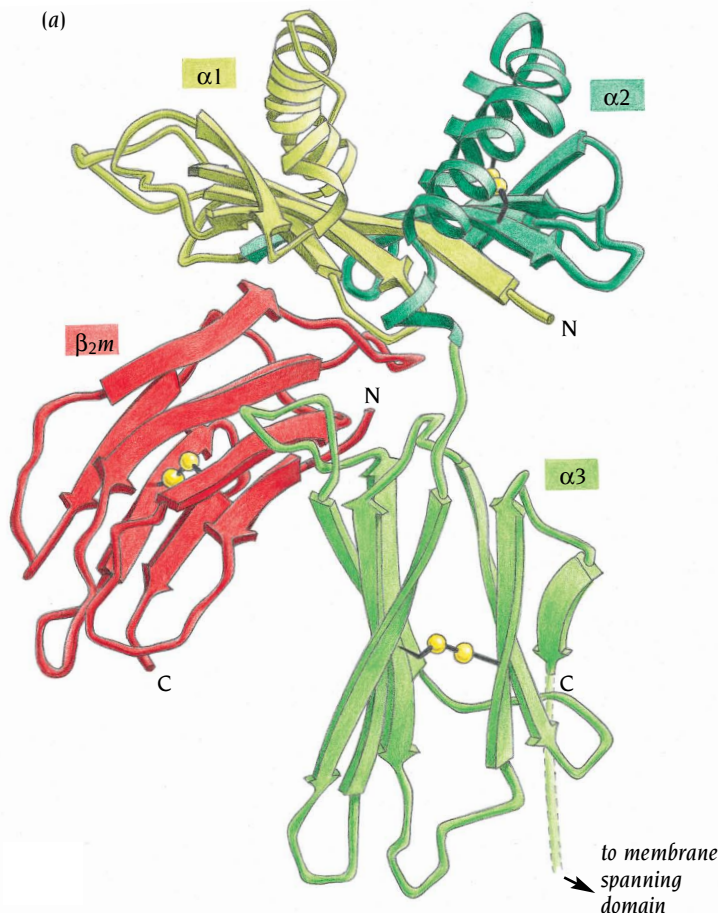
The answering of these questions began with the structure determination of the human class I MHC protein HLA-A2 (see Figure 15.18a). Subsequently, as methods became available for producing MHC proteins in recombinant expression systems and loading them *in vitro* with defined peptides, structures were determined for a class II MHC protein, and for single peptide complexes of both class I and class II proteins. These structures elegantly demonstrated that T cells recognize antigens as peptides bound in a groove in the MHC protein, and that peptide and MHC are recognized as a single surface. They have also helped to answer some important questions about MHC polymorphism and T-cell recognition and activation.

MHC molecules are composed of antigen-binding and immunoglobulin-like domains

Class I MHC molecules are plasma membrane proteins expressed in all cells and are composed of two polypeptide chains, a heavy chain of about 360 residues that spans the membrane bilayer and a noncovalently attached light chain of 99 residues. Class I MHC molecules present antigen to cytotoxic or “killer” T cells that destroy infected cells. The light chain, which is called β_2 microglobulin (β_2m), is homologous to the immunoglobulin constant domains and has essentially the same three-dimensional structure. The extracellular portion of the heavy chain is divided into three domains, α_1 , α_2 , and α_3 , each about 90 amino acids long. The carboxy-terminal domain α_3 also shows sequence and structural similarities to immunoglobulin constant domains, whereas α_1 and α_2 have very different structures. For structural analysis by x-ray crystallography, the extracellular portion of HLA-A2, composed of α_1 , α_2 , α_3 , and β_2m , was removed from the cells by proteolytic cleavage of a short flexible linker region that connects α_3 to the transmembrane domain.

The class I MHC structure is divided into two globular regions (Figure 15.18). One region is built up from the two immunoglobulin-like domains α_3

Figure 15.18 (a) Schematic representation of the path of the polypeptide chain in the structure of the class I MHC protein HLA-A2. Disulfide bonds are indicated as two connected spheres. The molecule is shown with the membrane proximal immunoglobulin-like domains (α_3 and β_2m) at the bottom and the polymorphic α_1 and α_2 domains at the top. (b) The domain arrangement in class I and class II MHC proteins. The domain structures of the MHC class II molecule are similar to those of the class I molecule shown in (a). [(a) Adapted from P.J. Björkman et al., *Nature* 329: 506–512, 1987.]



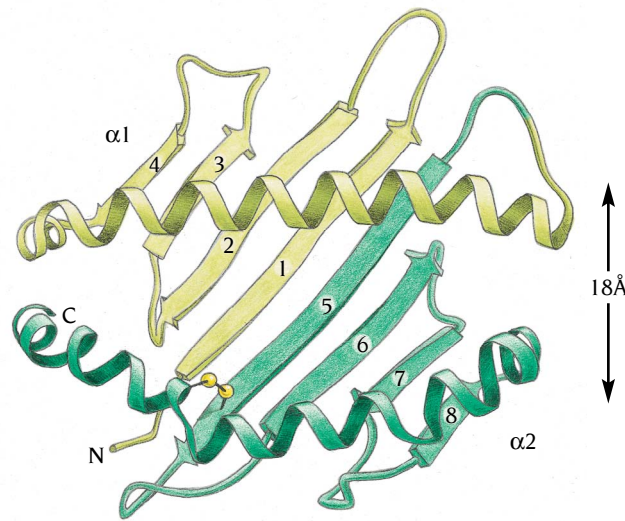


Figure 15.19 Schematic representation of the peptide-binding domain of a class I MHC protein. The $\alpha 1$ and $\alpha 2$ domains are viewed from the top of the molecule, showing the empty antigen-binding site as well as the surface that is contacted by a T-cell receptor. (Adapted from P.J. Björkman et al., *Nature* 329: 506–512, 1987.)

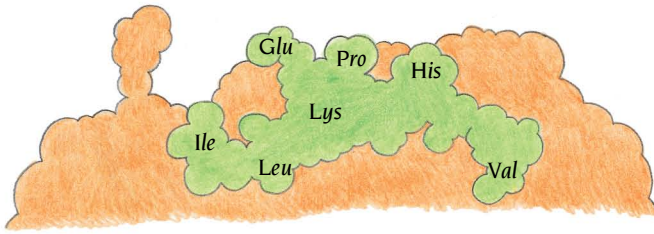
and β_2m . Since $\alpha 3$ is attached to the membrane in the intact HLA molecule, this region is closest to the cell surface. These domains are packed together through their four-stranded β sheets as for the pairing of the constant domains $C_{H1}-C_L$ and $C_{H3}-C_{H3}$ in IgG, but the relative orientation of the domains and the details of the packing are different. The second region, which contains the antigen binding site and presumably faces away from the membrane surface, is composed of domains $\alpha 1$ and $\alpha 2$. Each domain has a similar simple structure: starting from the N-terminus each region of the chain forms four up-and-down antiparallel β strands followed by a helical region across the β strands on one side of the β sheet. The two domains associate in such a way that their β sheets are hydrogen-bonded to each other in an antiparallel fashion. By this association, the structure of the complete region has a “floor” of a continuous eight-stranded antiparallel β sheet (Figure 15.19). This floor sits on top of and is stabilized by the immunoglobulin-like domains $\alpha 3$ and β_2m . Two helical regions are above the floor almost parallel to each other, and separated by a large distance, about 18 Å from center to center. A large crevice that faces the solution is thus formed with the floor forming its bottom and the helices its sides. This crevice is the antigen-binding site.

Recognition of antigen is different in MHC molecules compared with immunoglobulins

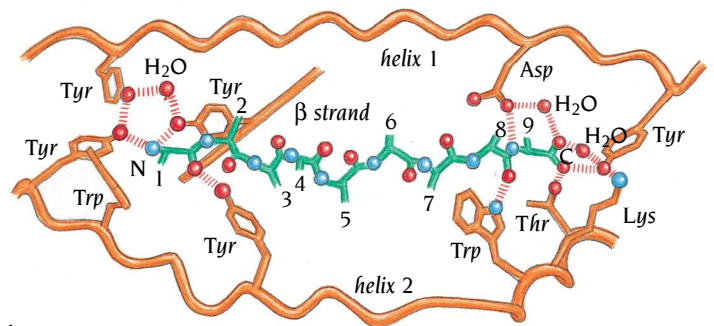
In the first structures determined for class I MHC proteins, the antigen binding site was occupied by a heterogeneous mixture of tightly bound peptides that copurified with the MHC. Subsequent structures for complexes with single, defined peptides have provided much information about the molecular details of peptide binding, which appears to be similar for all class I MHC proteins. Peptides bind with their N- and C-termini buried in the peptide-binding site, and their central regions exposed and bulging out from the cleft (Figure 15.20a). Sets of residues conserved in all class I MHC proteins form hydrogen bonds with these buried termini (Figure 15.20b). Peptides of 7 to 11 residues bind in this way to class I MHC, with varying conformations in the bulged region. Some of the side chains of the peptide are bound into small pockets or subsites within the overall binding cleft. These side chains are important to the overall peptide-binding affinity, while residues with side chains that project away from the MHC do not contribute substantially to the binding. In many cases the pockets accommodate the penultimate and the last residues of the peptide, but some class I MHC proteins have pockets into which central residues of the peptide bind.

The genes for MHC molecules, unlike immunoglobulin genes, do not undergo rearrangements to create structural diversity. The β_2m light chain is invariant, but the class I MHC heavy chain is the most genetically polymorphic

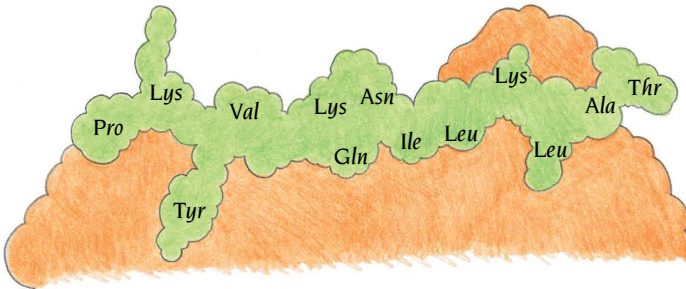
(a) class I



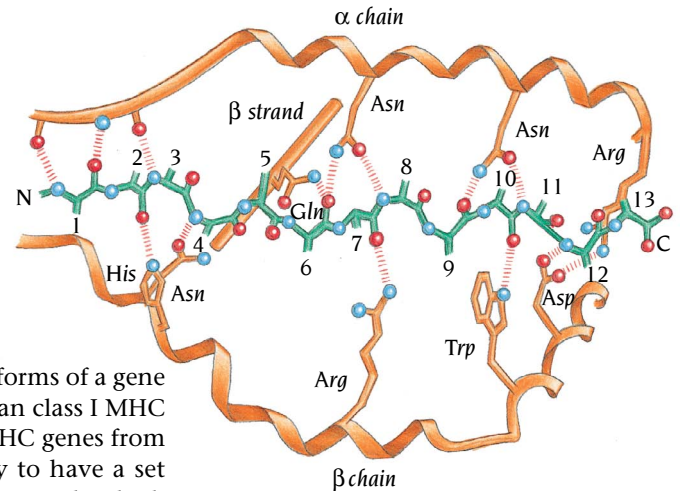
(b) class I



(c) class II



(d) class II



protein known, and more than one hundred alleles (alternate forms of a gene within the species) have been characterized for the three human class I MHC genes. Because of this polymorphism and the expression of MHC genes from both copies of chromosome 6, any given individual is likely to have a set of six different class I molecules that is almost unique to that individual. Differences among the MHC genes of different (mismatched) individuals account for transplantation rejection, and may contribute to differences in susceptibility to infection and autoimmune disease. Different MHC alleles have different binding preferences, as determined by analysis of the mixtures of endogenously bound peptides. How MHC gene polymorphism influences the selection of peptides by MHC proteins is clear from examination of MHC crystal structures: the polymorphic residues cluster in the side chain binding pockets.

Thus the mechanism of antigen recognition by MHC proteins differs from that of antibodies. Antibodies recognize a wide variety of different molecular species and conformations, and are sensitive to the three-dimensional structure of their antigen. MHC proteins recognize peptide antigens based on their sequence, and force the antigen into an extended conformation that displays many of the peptide side chains to the T-cell receptor (see below). Because MHC proteins almost always recognize peptides, some parts of their binding sites are conserved, while the pockets in the MHC molecule that bind the side chains of peptide antigens can vary between alleles to provide a range of binding specificity.

Peptides are bound differently by class I and class II MHC molecules

Class II MHC proteins are expressed on specialized antigen-presenting cells, such as macrophages or B cells, and are important in presenting antigen to helper T cells that regulate and coordinate the immune response to foreign antigen. Class II MHC proteins have the same basic tertiary structure as class I MHC, but the domains are distributed differently between the α and β subunits (see Figure 15.18b). In class II MHC, the subunits are of equal size, around 320 residues, and each chain folds into one immunoglobulin-like domain and one antigen-binding domain. Class II MHC proteins have a

Figure 15.20 Peptide binding by MHC class I and class II molecules. (a) A cross-section of a peptide antigen (green) bound to pockets in the class I molecule (orange). (b) Hydrogen bonds (red) form between the ends of the bound peptide and conserved residues of the class I molecule (orange). (c) Class II molecules bind longer peptides than class I molecules, with the ends of the peptide extending beyond the peptide-binding site. (d) Hydrogen bonds between bound peptide and the class II molecule occur along the length of the binding site, also in contrast to the case with class I complexes.

different binding specificity to class I MHC; they prefer to bind longer peptides (greater than 10 residues) with their ends extending from the binding site (see Figure 15.20c). The class II MHC molecule specifically recognizes side chains at more positions in the antigenic peptide than class I MHC, with strong preferences usually observed for positions 1, 4, 6, 9, where 1 refers to an arbitrary position near the N-terminus of the peptide (see Figure 15.20d).

This difference between class I and class II peptide-binding specificity appears to be due to small alterations and changes in the binding site. Additional conserved hydrogen-bonding residues are found in the center of the class II peptide-binding site, and the peptide does not bulge out from the MHC (compare Figures 15.20a and 15.20c). Interestingly, in all class II–peptide complexes determined to date, the peptide adopts a polyproline type II-like structure similar to polypeptides bound to SH3 domains as described in Chapter 13. This conformational homogeneity is in dramatic contrast to the conformational flexibility observed for peptides bound to class I MHC proteins.

The differences in peptide-binding specificity between short peptides with defined length for class I MHC and longer peptides for class II MHC may be related to the different mechanisms and intracellular locations for loading class I and class II MHC proteins with peptides. As membrane glycoproteins, both class I and class II MHC proteins are synthesized by membrane-bound ribosomes and assemble on the endoplasmic reticulum membrane. Class I MHC proteins generally bind peptides that have been derived from cytoplasmic proteins and transported into the endoplasmic reticulum by a dedicated transmembrane peptide pump (a mechanism used to alert the immune system to intracellular pathogens, for example, viruses that replicate in the cytoplasm). In contrast, class II MHC proteins move from the endoplasmic reticulum to a specialized endosomal compartment where they bind peptides derived from cell surface and extracellular proteins, a mechanism that detects extracellular or microsomal pathogens such as bacteria. These two mechanisms for peptide generation and loading, together with the ability of any particular MHC protein to bind many different peptide sequences, allow MHC molecules to display antigens derived from all types of cellular pathogens to the immune system.

T-cell receptors have variable and constant immunoglobulin domains and hypervariable regions

T-cell receptors (TCR) are heterodimeric transmembrane glycoproteins found exclusively in T cells, with extracellular domains that closely resemble antibody Fab structures. Each of the TCR α and β chains forms half of an extracellular antigen-binding domain, and in addition has one transmembrane

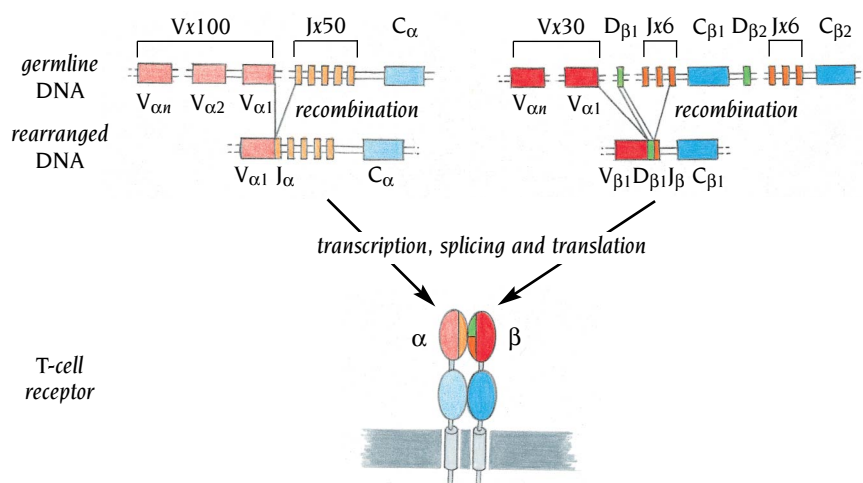


Figure 15.21 T-cell receptor α and β genes undergo rearrangements much like immunoglobulin genes (see Figure 15.5). The V α J α and V β D β J β regions correspond to the CDR loops. Nucleotides can also be added or deleted from the junctions, adding further amino acid sequence diversity.

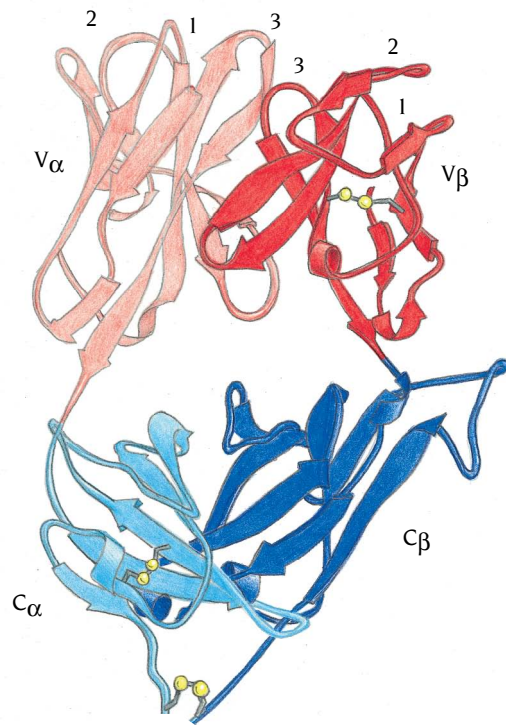


Figure 15.22 T-cell receptor structure shown as a ribbon diagram. The antigen-binding site is formed by CDR loops (labeled 1 to 3) from the V_α and V_β domain, as for antibodies. A disulfide bond (yellow) links the two peptide chains. (Courtesy of A.I. Wilson.)

domain and a short cytoplasmic tail. Like the antibody light chains, TCR α and β chains both have one variable and one constant domain as well as hypervariable regions that are involved in antigen recognition. In the T-cell plasma membrane, TCR assembles with other proteins to form a signaling complex that transmits information across the T-cell membrane in response to antigen binding. TCR $\alpha\beta$ genes are rearranged during T cell development much as are antibody genes, except that the number of J genes is larger, and nucleotides are added to both V_α - D_α - J_α and V_β - J_β junctions (Figure 15.21). Unlike antibody genes, however, TCR genes are not subject to somatic mutation after rearrangement.

In 1996, crystal structures were determined for a murine TCR (by the group of Ian Wilson at Scripps Institute) and for a human TCR-MHC-peptide complex (by the group of Don Wiley at Harvard University), which together with earlier structures for isolated TCR α and β chains (by Roy Mariuzza and coworkers, University of Maryland), provided a coherent view of TCR structure. As predicted much earlier by Cyrus Chothia on the basis of homology with immunoglobulins, each TCR chain forms two immunoglobulin-like domains (Figure 15.22). The membrane proximal constant domains C_α and C_β are similar to C_{H1} , C_{H2} , and C_{H3} , and the membrane distal variable domains V_α and V_β are similar to V_H and V_L . Packing between the V_α and V_β domains and between the C_α and C_β domains is similar to that found in antibody Fab fragments. A single disulfide bond at the end of the protein links the C_α and C_β domains, and is equivalent to the bond that links C_{H1} with C_L . Finally, the antigen combining site is formed by hypervariable CDR loops in the variable domains, as it is in antibodies.

In spite of these similarities, some alterations to the basic immunoglobulin fold are found in the TCR subunits and these may be important for function. The C'' strand in the V_β domain is switched from the inner G-F-C-C' sheet to the outer A-B-E-D sheet, resulting in a different shape for the CDR2 loop. There is also an unusual insertion in the C_β F-G connection that places a large loop near the $V_\beta C_\beta$ interface, which extends away from the protein surface (see the right edge of Figure 15.22). This loop may well be important in interactions with other components of the overall TCR complex. Finally, the C_α domain is unusual and structurally divergent from other Ig domains.

Figure 15.23 Interactions between the class I MHC–peptide complex with the T-cell receptor. (a) The T-cell receptor (top) binds to the MHC–peptide complex with its hypervariable CDR loops effectively burying the eight-residue foreign peptide (yellow, with the N- and C-terminal residues labeled P1 and P8, respectively). The TCR α subunit loops 1–3 are colored light purple (CDR1 α), dark purple (CDR2 α), and yellow (CDR3 α); the β chain loops 1–3 are colored light blue (CDR2 β), dark blue (CDR3 β), and green (CDR3 β). The β subunit hypervariable loop that has no counterpart in immunoglobulins, HV4, is shown in red. (b) A finger-print of how the TCR hypervariable loops contact the MHC–peptide complex, as seen looking down onto the MHC surface as oriented in (a). The TCR sits diagonally across the peptide-binding site, contacting both the MHC molecule (gray and green space-filling model) and its bound peptide (pale green space-filling model). (Courtesy of A.I. Wilson.)

MHC–peptide complexes are the ligands for T-cell receptors

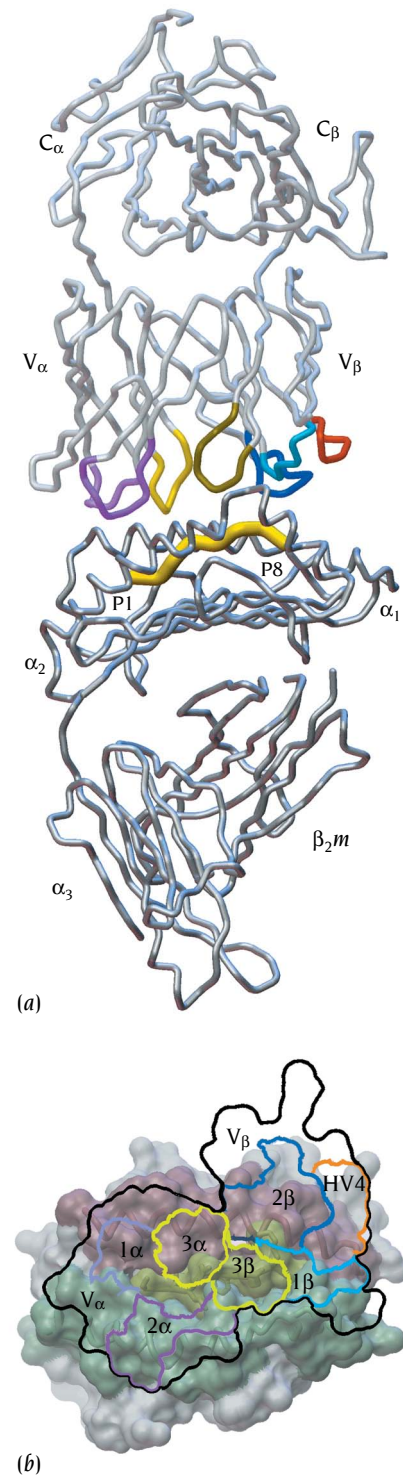
The physiological role of TCR is to bind MHC–peptide complexes presented by antigen-presenting cells. During T cell development, cells bearing TCR that bind to MHC complexed with self peptides are removed from the immune system; thus mature T cells carry only receptor that can bind to MHC proteins with foreign antigens bound in their peptide binding groove. The structures of several class I MHC–peptide complexes bound to TCR $\alpha\beta$ have been determined. These all show the MHC and TCR proteins interacting diagonally, with the TCR contacting the peptide and both helices of the MHC protein such that the peptide is virtually buried in the interface (Figure 15.23a).

In the TCR, packing of the V_α and V_β domains brings together CDR1, CDR2, and CDR3 to form a relatively flat combining site well-suited to interact with the surface of the MHC–peptide complex. Among these three loops, CDR3 is the largest and most variable as it is in immunoglobulins, and appears strongly to determine peptide specificity. In the TCR–MHC–peptide complex, the CDR loops lie over the MHC peptide-combining site, with the CDR3 loops from V_α and V_β (CDR3 α and CDR3 β) packed together over the center of the bound peptide (see Figure 15.23b). CDR1 α and CDR2 α extend diagonally towards the N-terminal end of peptide, and also contact MHC residues from the α_2 subunit helix. CDR1 β and CDR2 β are oriented correspondingly toward the peptide's C-terminus and the MHC α_1 subunit helix. Thus, all the CDR loops of the TCR can potentially contact the MHC molecule and/or bound peptide. Small changes in the CDR loops of the receptor, and in exposed regions of the peptide and MHC protein, have been observed upon complex formation, but no large conformational changes that could account for signal transduction have been seen.

To date, there has not been a reported structure determination of a class II MHC–peptide–TCR complex. However, T-cell receptors that recognize class I MHC–peptide complexes and those that recognize class II MHC–peptide complexes utilize the same set of V_α and V_β genes, and the principal feature that defines the site of class I MHC–TCR interaction, the cleft formed by the α_1 and α_2 subunits of the MHC molecule, is shared by both class I and class II proteins (see Figure 15.20b and c). Thus, the general mode of class II MHC interaction with TCR is expected to be similar to that determined for class I MHC.

Many cell-surface receptors contain immunoglobulin-like domains.

Many other cell-surface proteins involved in immunological recognition utilize immunoglobulin-like domains as structural elements. Immunoglobulin domains have been classified into five types, namely V (like antibody variable



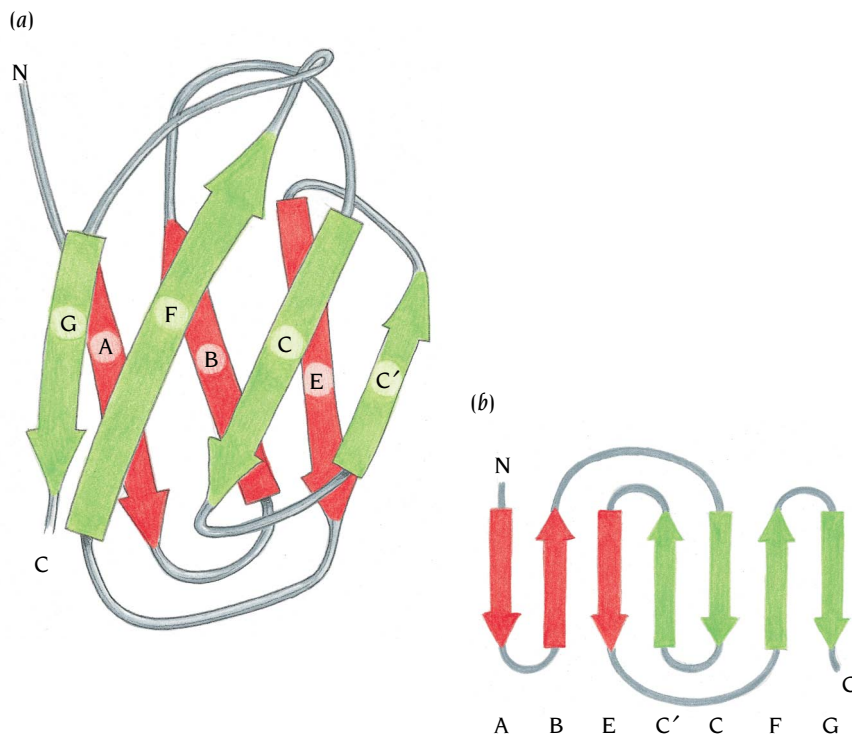


Figure 15.24 Ribbon diagram (a) and topology diagram (b) of the fibronectin type III domain, which is composed of a three-stranded and a four-stranded β sheet packed together as a compressed barrel.

domains), C1 (like antibody constant domains) C2 (strand-switched variant of C1), I, and E, based on structural and sequence homology. A close cousin of the immunoglobulin domain is the fibronectin type III domain, which has a β -sandwich fold closely related to the immunoglobulin fold (Figure 15.24). Hundreds of proteins appear to be made up of combinations of these domains in an essentially modular fashion. Interaction sites on immunoglobulin domains are sometimes found in the CDR loops or equivalent regions, but can also involve other parts of the domain. In many cases, the immunoglobulin domains appear to play a structural role not directly involved in ligand binding.

An interesting adaptation is found in CD4, a T-cell coreceptor that binds class II MHC along with the TCR during T cell activation. In addition to its role in T cell activation, CD4 also serves as the major binding site for the human immunodeficiency virus (HIV), through interactions with the HIV surface glycoprotein receptor, gp120. CD4 is a monomer with four extracellular Ig domains. The structure of the first and second extracellular domains, which include the binding sites for MHC and for gp120, was determined simultaneously in 1990 by the groups of Stephen Harrison at Harvard University, and Wayne Hendrickson at Columbia University. The first domain is a V-type immunoglobulin with strand A completely switched to the G-F-C-C'-C'' sheet: the second domain is a conventional C2 type. At the junction of the first and second domains, the last strand from one domain continues uninterrupted to form the first strand of the next domain (Figure 15.25). Presumably this arrangement provides rigidity to the molecule. A similar arrangement has also been observed between the third and fourth domains of CD4. Sharing β strands between domains is one way to confer rigidity to a modular protein.

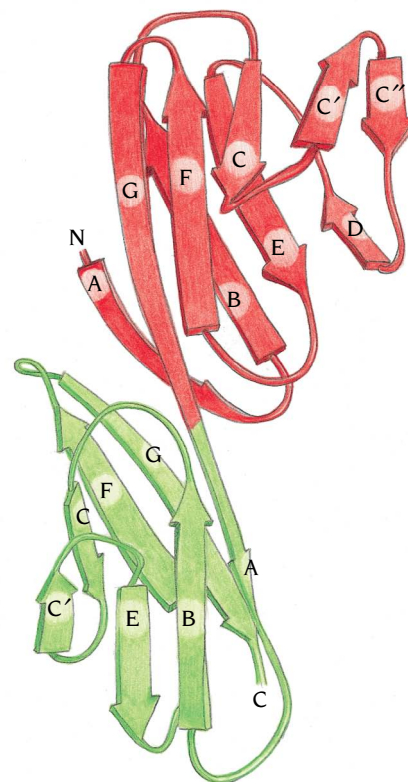


Figure 15.25 Ribbon diagram for the first two domains of CD4. The G strand of domain 1 is contiguous with the A strand of domain 2. The C'' ridge has been implicated in binding to the HIV surface protein, gp120.

Conclusion

IgG antibody molecules are composed of two light chains and two heavy chains joined together by disulfide bonds. Each light chain has one variable domain and one constant domain, while each heavy chain has one variable and three constant domains. All of the domains have a similar three-dimensional structure known as the immunoglobulin fold. The Fc stem of the molecule is formed by constant domains from each of the heavy chains, while two Fab arms are formed by constant and variable domains from both heavy and light chains. The hinge region between the stem and the arms is flexible and allows the arms to move relative to each other and to the stem.

The constant domain has a stable framework structure composed of two antiparallel sheets comprising seven β strands, four in one sheet and three in the other. The variable domains have a similar framework structure but comprising nine β strands, five in one sheet and four in the other. The three hypervariable regions are in loops at one end of the variable domain. The variable domains from the heavy and light chains associate through their five-stranded β sheets to form a barrel with the hypervariable loop regions from both domains close together at the top of the barrel.

By this arrangement the hypervariable loops can form antigen-binding sites of different shapes depending on the size and sequence of the loops. Haptens induce antibodies with binding cavities, and protein antigens induce antibodies with extended binding surfaces, which contain grooves or pockets for protruding antigen residues. In general, no gross structural alterations occur in the antibody or antigen when they bind together, although a variety of small conformational changes may be seen. The third hypervariable region, CDR3, of the heavy chain is in the center of the interaction area and makes a large number of contacts with the antigen. This region corresponds to the D gene segment, which, through junctional diversity and addition of nucleotides during the recombination events, has the largest sequence variation of the hypervariable regions.

Class I and class II MHC molecules bind peptide antigens and present them at the cell surface for interaction with receptors on T cells. The extracellular portion of these molecules consists of a peptide-binding domain formed by two helical regions on top of an eight-stranded antiparallel β sheet, separated from the membrane by two lower domains with immunoglobulin folds. These domains are differently disposed between the two protein subunits in class I and class II molecules.

Class I MHC molecules bind short peptides of 8–11 residues, with the peptide's termini buried within the protein and the central regions exposed for interactions with the T-cell receptor. Conserved residues interact with main chain groups of the peptide's termini, and two or three peptide side chains at particular positions bind into pockets within the overall binding groove. These pockets vary between MHC class I alleles, and define the peptide-binding specificity for different class I MHC molecules. Class II MHC molecules bind longer peptides, the ends of which are exposed and extend beyond the peptide-binding pocket. Conserved residues in the MHC molecules interact with main-chain atoms along the peptide. Class II peptide-binding motifs generally bind their ligands at more positions than those of class I. Both class I and class II MHC proteins are extremely polymorphic, with hundreds of allelic variants for some MHC genes.

T-cell receptors are cell surface proteins with extracellular domains composed of two similarly sized subunits, each with a variable and a constant immunoglobulin-like domain, arranged similarly to the Fab arms of antibodies. Like antibodies, T-cell receptors have an antigen-binding site formed by hypervariable loop regions from both variable domains, with the CDR3 loops playing a key role in antigen recognition. Unlike antibodies, T-cell receptors recognize antigen only when it is bound in the peptide-binding site of an MHC protein. In ternary complexes of T-cell receptor, MHC molecule, and

peptide antigen that have been determined to date, the T-cell receptor lies diagonally across the top of the MHC peptide-binding site, and contacts exposed regions of the peptide as well as the MHC helices.

The immune system can use a few different types of protein modules to bind to an almost infinite number of antigens, achieving binding diversity by using variable loops or pockets supported by a constant framework. The immunoglobulin-like fold often forms the basis of this framework. If a particular type of antigen is to be recognized, the binding site will contain conserved regions that contact invariant portions of the antigen; if the antigens are diverse, the binding regions will be variable. For antibodies and T-cell receptors, the required variability exceeds the number of genes in the genome, and gene recombination and mutation are utilized to generate diversity.

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The Structure of Spherical Viruses

16

All viruses depend for their existence on their ability to infect cells, causing them to make more virus particles. The infected cells generally die in the process. Even though the infection cycles of different types of viruses vary in detail, they all follow the same basic pattern. First, the viruses deliver their nucleic acid genome into the host cell, often in association with viral proteins; the mechanism of delivery varies depending on the virus, but the end result is the same. Second, the host cell's biosynthetic machinery is subverted for the replication, transcription, and translation of the viral genes at the expense of cellular gene expression. Finally, progeny virus particles assemble in the infected cell and by one route or another leave it to infect a fresh host cell. Individual viruses can only infect a restricted range of hosts, and the host organisms usually have defense mechanisms, ranging from restriction enzymes in bacteria to the immune system of vertebrates. The combination of restricted host ranges and host defense mechanisms keeps in check the extent of viral infections.

Viruses come in many different sizes, shapes, and compositions. They all have a nucleic acid genome, but this is DNA in some viruses and RNA in others, double stranded in some but single stranded in others, a single molecule in some but segmented into several different molecules in others. The nucleic acid genome is always surrounded by a protein shell, a **capsid**. Some viruses also have a lipid bilayer membrane, an **envelope**, enclosing the capsid; one example is the influenza viruses discussed in Chapter 5. For animal viruses the distinction between nonenveloped and enveloped viruses is important because it corresponds to different strategies for entry into and exit from a cell. Enveloped viruses bud out through a cellular membrane, acquiring the lipid bilayer in the process. Their entry into cells involves a membrane fusion event. Nonenveloped viruses have evolved more complex strategies and some have elaborately differentiated structures. For example, bacteriophage T4, which must penetrate the bacterial cell wall at specific sites and inject its DNA into the cell, is seen in electron micrographs to resemble a hypodermic syringe.

A nucleic acid can never code for a single protein molecule that is big enough to enclose and protect it. Therefore, the protein shell of viruses is built up from many copies of one or a few polypeptide chains. The simplest viruses have just one type of capsid polypeptide chain, which forms either a rod-shaped or a roughly spherical shell around the nucleic acid. The simplest such viruses whose three-dimensional structures are known are plant and insect viruses: the rod-shaped tobacco mosaic virus, the spherical satellite tobacco necrosis virus, tomato bushy stunt virus, southern bean mosaic virus,

and flock house virus. The mammalian parvoviruses also have a simple, single-subunit design. The mammalian picornaviruses have a spherical shell built up of four different polypeptide chains. The structures of several picornaviruses are known: common cold virus, Mengo virus, poliovirus, and foot-and-mouth disease virus. The largest virus particles whose structures have been determined are SV40 and polyomavirus, which are closely related double-stranded DNA viruses, and the inner capsid particle of bluetongue virus, a double-stranded RNA virus. These viruses all infect species of mammals. Schematic diagrams illustrating the size and shape of some of the viruses mentioned above are given in Figure 16.1.

The protein shells, or capsids, of the simple spherical animal and plant viruses are made up of symmetrically arranged protein subunits. This symmetry of subunits has been exploited by crystallographers to facilitate the determination of the structure of the capsids at atomic resolution by x-ray methods. Consequently, these viruses are the largest aggregates of biological macromolecules whose structures have been determined at high resolution. The nucleic acid genome, which is inside the protein shell of the virus, lacks symmetry and is unlikely to have a unique structure. However, we are beginning to understand packaging signals and the unique protein-RNA or protein-DNA interactions that initiate virus particle assembly. As we will see later, the unique interactions in the interior of an RNA phage have been seen crystallographically by making a complex of the MS2 capsid with 90 identical RNA fragments.

Detailed knowledge of the structure of viral capsids alone is valuable for both practical and theoretical reasons. In principle, it should be possible to exploit this information for the rational design of antiviral drugs. For instance, the entry of mammalian viruses into cells depends upon specific recognition by the viral capsid of proteins or carbohydrates that occur on the surface of mammalian cells. This interaction thus plays a crucial part in determining the viral host range, and compounds that interfere with it should block infection. Understanding the capsid structure should help in designing such compounds. Because viral binding sites for surface molecules of host cells are also a target for protective antibodies, the immune system of the host has in all probability played a major role in the evolution of animal viruses. Detailed knowledge of the surface regions of these viruses should therefore teach us more about the structural adaptations that viruses have evolved to evade host defences.

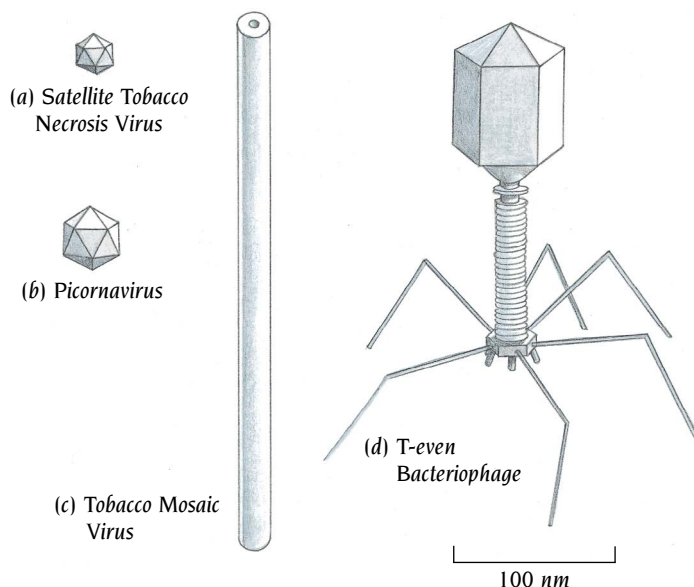


Figure 16.1 Viruses vary in size and shape from the simplest satellite viruses (a) that need another virus for their replication to the T-even bacteriophages (d) that have developed sophisticated mechanisms for injecting DNA into bacteria. Four different virus particles are shown to scale.

In this chapter we will examine the construction principles of spherical viruses, the structures of individual subunits and the host cell binding properties of the surface of one of the picornaviruses, the common cold virus.

The protein shells of spherical viruses have icosahedral symmetry

Two basic principles govern the arrangement of protein subunits within the shells of spherical viruses. The first is specificity; subunits must recognize each other with precision to form an exact interface of noncovalent interactions because virus particles assemble spontaneously from their individual components. The second principle is genetic economy; the shell is built up from many copies of a few kinds of subunits. These principles together imply symmetry; specific, repeated bonding patterns of identical building blocks lead to a symmetric final structure.

The first question then is, what types of symmetry can form roughly spherical objects that are built up from identical units. In other words, how can one make a sphere by arranging identical objects symmetrically on its surface?—a problem also faced by producers of soccer balls. This question was answered more than 2000 years ago by the classical Greek mathematicians, who showed that there is only a very limited number of ways to build such objects. Among these, the icosahedron and dodecahedron have the highest possible symmetry, generally called **icosahedral symmetry**, and therefore allow a maximum number of identical objects to form a closed symmetrical shell (Figure 16.2). Do these geometric principles apply to viral structures?

Fortunately, virus particles are large enough to be studied by electron microscopy using negative staining. In the 1950s and 1960s such studies, mainly by Aaron Klug and his collaborators in Cambridge, England, showed that indeed all the spherical viruses examined exhibited icosahedral symmetry. They could also demonstrate from the x-ray diffraction patterns of virus crystals that not only does the external surface obey this symmetry, but so also does the arrangement of subunits that make up this surface. Since icosahedral symmetry is central to an understanding of the architecture of spherical viruses, it will be examined in detail.

The icosahedron has high symmetry

The icosahedron is a roughly spherical object that is built up from 20 identical equilateral triangles. These triangular tiles are arranged side by side in such a way that they enclose the volume inside the icosahedron. Figure 16.3 shows a view of an icosahedron where there are 5 tiles at the top, 5 at the bottom, and 10 in a band around the middle region. These faces are identical: in other words, identical icosahedra are obtained irrespective of which of the vertices is at the top in Figure 16.3a.

The symmetry of the icosahedron is described by the different types of rotations that bring it into self-coincidence (see Figure 16.3). There are 12 vertices, each of which has a fivefold rotation axis that passes through it and through the center. Rotation by a fifth of a revolution about one of these axes brings the icosahedron to an orientation indistinguishable from the starting orientation. In addition, there are 20 threefold symmetry axes through the center of the icosahedron and the middle of each of the 20 faces, and 30 twofold axes through the middle of each of the 30 edges. All this symmetry has important consequences for the way in which the shell of spherical viruses are constructed.

Any symmetric object is built up from smaller pieces that are identical and that are related to each other by symmetry. An icosahedron can therefore be divided into a number of smaller identical pieces called symmetry-related units. Protein subunits are asymmetric objects; hence, a symmetry axis cannot pass through them. The minimum number of protein subunits that can form a virus shell with icosahedral symmetry is therefore equal to

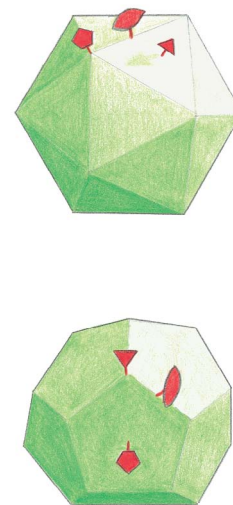


Figure 16.2 The icosahedron (*top*) and dodecahedron (*bottom*) have identical symmetries but different shapes. Protein subunits of spherical viruses form a coat around the nucleic acid with the same symmetry arrangement as these geometrical objects. Electron micrographs of these viruses have shown that their shapes are often well represented by icosahedra. One each of the twofold, threefold, and fivefold symmetry axes is indicated by an ellipse, triangle, and pentagon, respectively.

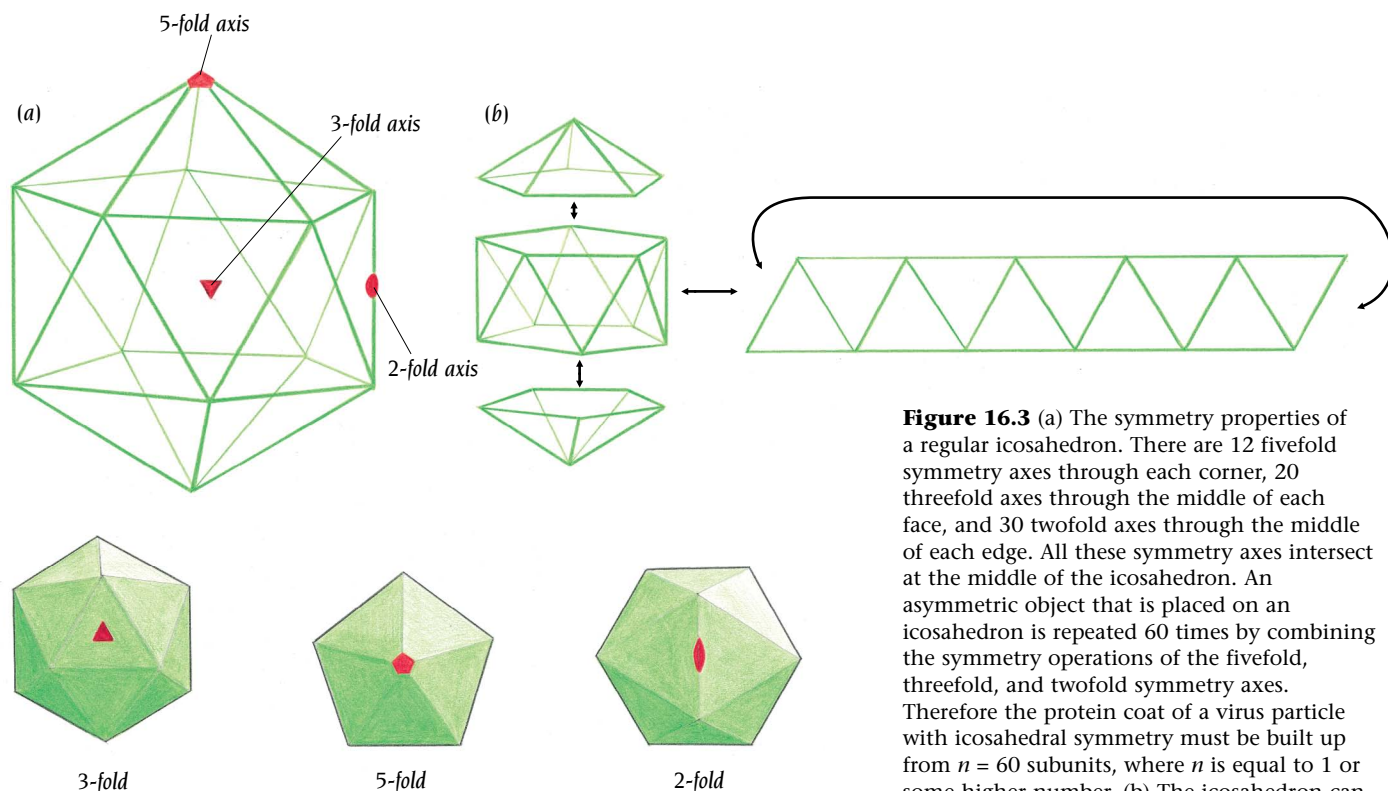


Figure 16.3 (a) The symmetry properties of a regular icosahedron. There are 12 fivefold symmetry axes through each corner, 20 threefold axes through the middle of each face, and 30 twofold axes through the middle of each edge. All these symmetry axes intersect at the middle of the icosahedron. An asymmetric object that is placed on an icosahedron is repeated 60 times by combining the symmetry operations of the fivefold, threefold, and twofold symmetry axes. Therefore the protein coat of a virus particle with icosahedral symmetry must be built up from $n = 60$ subunits, where n is equal to 1 or some higher number. (b) The icosahedron can be regarded as being built up of 5 identical equilateral triangular tiles at the top, 5 at the bottom, and 10 in a band around the middle region. (c) The icosahedron viewed along each of its different symmetry axes.

the number of symmetry-related units in an icosahedron. This number can easily be deduced by regarding the 20 triangular tiles on the surface of the icosahedron. These tiles have threefold symmetry, and therefore, by definition, three identical objects are required to form one tile. One obvious way to regard such tiles is shown in Figure 16.4a, which illustrates the division of one such triangle into three equal parts, each of which contains an object. Since there are 20 tiles the total number of units in the icosahedron is $20 \times 3 = 60$. We could equally well have regarded the 12 different corners with fivefold symmetry or the 30 edges with twofold symmetry. Such units are called **asymmetric units**; thus the icosahedron and, by inference, spherical viruses have 60 asymmetric units (Figure 16.4b). Hence at least 60 protein subunits are required to form a virus shell with icosahedral symmetry.

The symmetry properties of an icosahedron are not restricted to the surface but extend through the whole volume. An asymmetric unit is therefore a part of this volume; it is a wedge from the surface to the center of the icosahedron. Sixty such wedges completely fill the volume of the icosahedron.

The simplest virus has a shell of 60 protein subunits

The asymmetric unit of an icosahedron can contain one or several polypeptide chains. The protein shell of a spherical virus with icosahedral symmetry

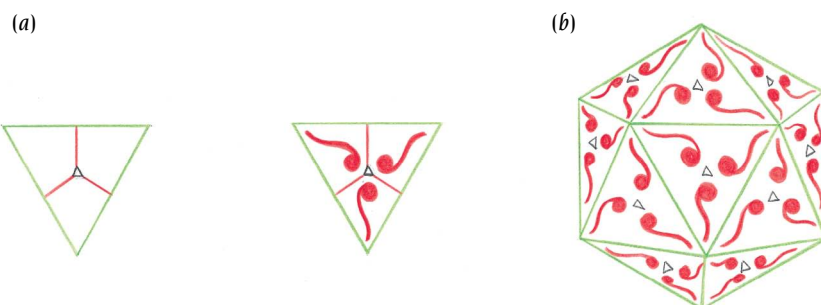


Figure 16.4 The division of the surface of an icosahedron into asymmetric units. (a) One triangular face is divided into three asymmetric units into which an object is placed. These are related by the threefold symmetry axis. (b) Triangular tiles with the three symmetry-related objects have been positioned on the surface of the icosahedron in such orientations that the objects are related by the fivefold and twofold symmetry axes.

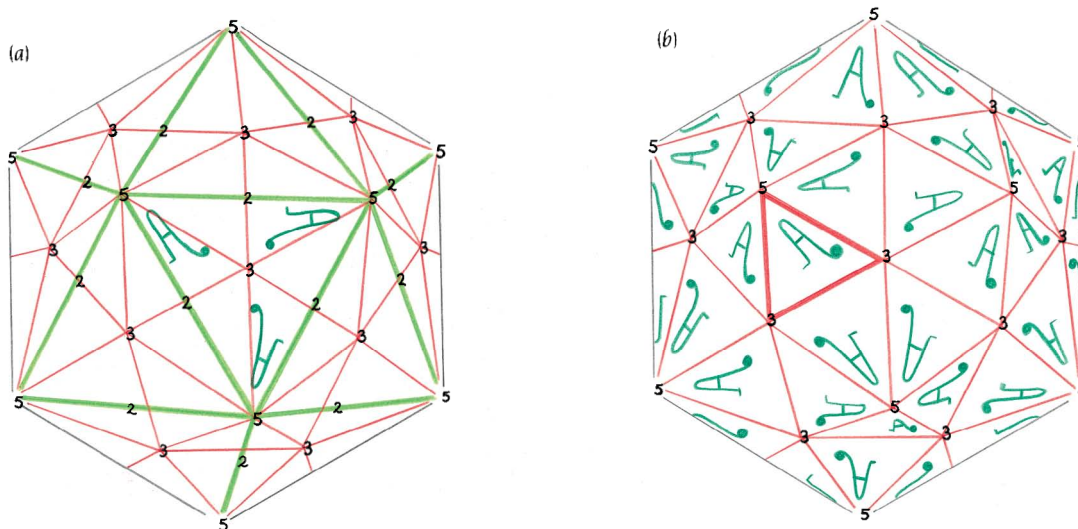


Figure 16.5 Schematic illustration of the way the 60 protein subunits are arranged around the shell of satellite tobacco necrosis virus. Each subunit is shown as an asymmetric A. The view is along one of the threefold axes, as in Figure 16.3a. (a) Three subunits are positioned on one triangular tile of an icosahedron, in a similar way to that shown in 16.4a. The red lines represent a different way to divide the surface of the icosahedron into 60 asymmetric units. This representation will be used in the following diagrams because it is easier to see the symmetry relations when there are more than 60 subunits in the shells. (b) All subunits are shown on the surface of the virus, seen in the same orientation as 16.4a. The shell has been subdivided into 60 asymmetric units by the red lines. When the corners are joined to the center of the virus, the particle is divided into 60 triangular wedges, each comprising an asymmetric unit of the virus. In satellite tobacco necrosis virus each such unit contains one polypeptide chain.

must therefore have a minimum number of 60 identical units. Such a unit can be a single protein chain, a dimer of two identical protein chains as in the inner shell of bluetongue virus or several different protein chains as in the picornaviruses to be described later. Is there any virus known with only 60 protein chains, or is the volume inside such a shell too small to enclose the viral genome?

Very few self-sufficient viruses have only 60 protein chains in their shells. The satellite viruses do not themselves encode all of the functions required for their replication and are therefore not self-sufficient. The first satellite virus to be discovered, **satellite tobacco necrosis virus**, which is also one of the smallest known with a diameter of 180 Å, has a protein shell of 60 subunits. This virus cannot replicate on its own inside a tobacco cell but needs a helper virus, tobacco necrosis virus, to supply the functions it does not encode. The RNA genome of the satellite virus has only 1120 nucleotides, which code for the viral coat protein of 195 amino acids but no other protein. With this minimal genome the satellite viruses are obligate parasites of the viruses that parasitize cells.

The structure of satellite tobacco necrosis virus has been determined to 2.5 Å resolution by the group of Bror Strandberg in Uppsala, Sweden. As expected, the viral capsid has icosahedral symmetry with one polypeptide chain in the asymmetric unit. This is schematically illustrated in Figure 16.5a. The subunits are arranged so that they are quite close together around the fivefold axes forming tight contacts at the surface of the particle (Figure 16.5b). On the inside of the shell, on the other hand, the polypeptide chains are clustered around the threefold axes, one α helix from each subunit forming a bundle of three α helices. These tight interactions around the fivefold and threefold axes form a scaffold that links all subunits together to complete the shell.

The outer diameter of the shell is approximately 180 Å and the inner diameter about 125 Å except around the threefold axes, where the N-terminal α helices project about 20 Å into the core. The RNA molecule that is present in the core is not visible in the electron density map.

Complex spherical viruses have more than one polypeptide chain in the asymmetric unit

We have seen in the structure of this simple satellite virus that 60 subunits are sufficient to form a shell around an RNA molecule that codes for the subunit protein, but there is little room for additional genetic information.

Self-sufficient viruses have longer genomes, which code for enzymes essential for the replication of the viral nucleic acid in addition to the structural proteins of the capsid. These genomes need a larger volume inside the capsid and therefore need a larger capsid. How are such larger shells constructed within the constraints of icosahedral symmetry? Very little might be gained by increasing the size of each subunit because the shell could become thicker as it is enlarged. The only way out is to increase the number of subunits. To preserve icosahedral symmetry, the asymmetric unit itself must then contain more than one subunit. Since there are 60 asymmetric units, the total number of subunits in the shell must therefore be a multiple of 60. These subunits can be either identical or different. Genetic economy is achieved by using many identical subunits; only one gene is needed to code for the entire protein shell. There are many examples of viruses of this type, and we will examine one example before describing viruses that contain different types of subunits in their capsids.

Can any number of identical subunits be accommodated in the asymmetric unit while preserving specificity of interactions within an icosahedral arrangement? This question was answered by Don Caspar then at Children's Hospital, Boston, and Aaron Klug in Cambridge, England, who showed in a classical paper in 1962 that only certain multiples (1, 3, 4, 7...) of 60 subunits are likely to occur. They called these multiples **triangulation numbers, T**. Icosahedral virus structures are frequently referred to in terms of their triangulation numbers; a $T = 3$ virus structure therefore implies that the number of subunits in the icosahedral shell is $3 \times 60 = 180$.

Caspar and Klug based their arguments on the principle of specificity, or that the protein-protein contacts should be approximately similar. When T is larger than one, it is no longer possible to pack the protein subunits into an icosahedral shell in a strictly equivalent way as it is for a $T = 1$ structure like satellite tobacco necrosis virus where all subunits have the same environment and hence the same packing interactions. What Caspar and Klug showed is that for certain specific values of T above one (3, 4, 7...) it is possible to pack the subunits with only slightly different bonding patterns, in a **quasi-equivalent** way.

As examples of such quasi-equivalent arrangement of subunits, we will examine the $T = 3$ and $T = 4$ packing modes, both of which are found in known virus particles. In the $T = 3$ structure, which has 180 subunits (3×60),

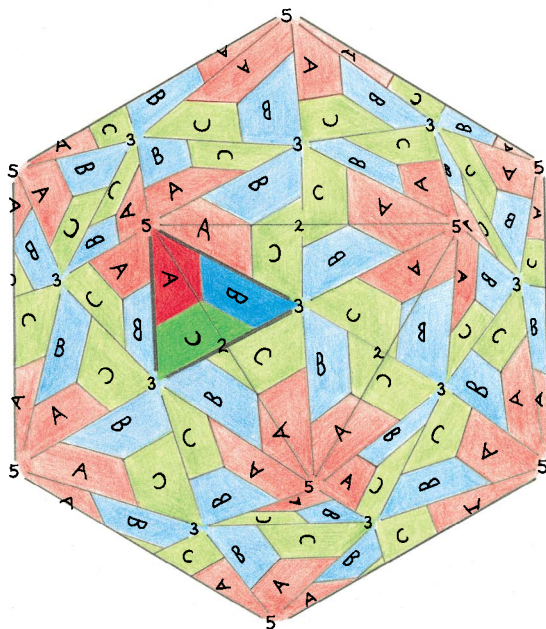


Figure 16.6 A $T = 3$ icosahedral virus structure contains 180 subunits in its protein shell. Each asymmetric unit (one such unit is shown in thick lines) contains three protein subunits A, B, and C. The icosahedral structure is viewed along a threefold axis, the same view as in Figure 16.5. One asymmetric unit is shown in dark colors.

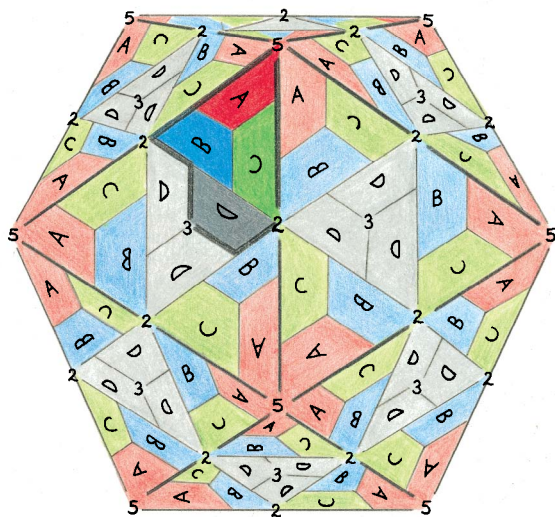


Figure 16.7 A $T = 4$ icosahedral virus structure contains 240 subunits in its protein shell. Each asymmetric unit (one such unit is shown in dark colors) contains four protein subunits, A, B, C, and D. We have now twisted the icosahedron somewhat, compared with Figure 16.6, so that we view the structure along one of the twofold axes, the same view as in Figure 16.3c.

each asymmetric unit contains three protein subunits with different environments, which we will call A, B, and C. These are arranged so that the A subunits interact around the fivefold axes, and the B and C subunits alternate around the threefold axes (Figure 16.6). There are thus six subunits, three each of B and C, arranged in a pseudosymmetric way around the threefold axes, which therefore are also pseudosixfold axes. Caspar and Klug argued that the subunits could pack around a fivefold and a pseudosixfold axis in a rather similar fashion with only minor alterations of the packing mode. We will see in the structure of tomato bushy stunt virus that they were right. This viral capsid is made from chemically identical proteins that are able to accommodate the necessary differences in packing mode by changing their conformation according to their different environments.

In the $T = 4$ structure there are 240 subunits (4×60) in four different environments, A, B, C, and D, in the asymmetric unit. The A subunits interact around the fivefold axes, and the D subunits around the threefold axes (Figure 16.7). The B and C subunits are arranged so that two copies of each interact around the twofold axes in addition to two D subunits. For a $T = 4$ structure the twofold axes thus form pseudosixfold axes. The A, B, and C subunits interact around pseudothreefold axes clustered around the fivefold axes. There are 60 such pseudothreefold axes. The $T = 4$ structure therefore has a total of 80 threefold axes: 20 with strict icosahedral symmetry and 60 with pseudosymmetry.

Structural versatility gives quasi-equivalent packing in $T = 3$ plant viruses

The molecular basis for quasi-equivalent packing was revealed by the very first structure determination to high resolution of a spherical virus, **tomato bushy stunt virus**. The structure of this $T = 3$ virus was determined to 2.9 Å resolution in 1978 by Stephen Harrison and co-workers at Harvard University. The virus shell contains 180 chemically identical polypeptide chains, each of 386 amino acid residues. Each polypeptide chain folds into distinct modules: an internal domain R that is disordered in the structure, a region (a) that connects R with the S domain that forms the viral shell, and, finally, a domain P that projects out from the surface. The S and P domains are joined by a hinge region (Figure 16.8).

When they form the three subunits A, B, and C of the asymmetric unit, the identical polypeptides adopt different three-dimensional structures. The C subunit in particular is distinct from the A and B structures, its hinge region assuming a different conformation so that the S and P domains are

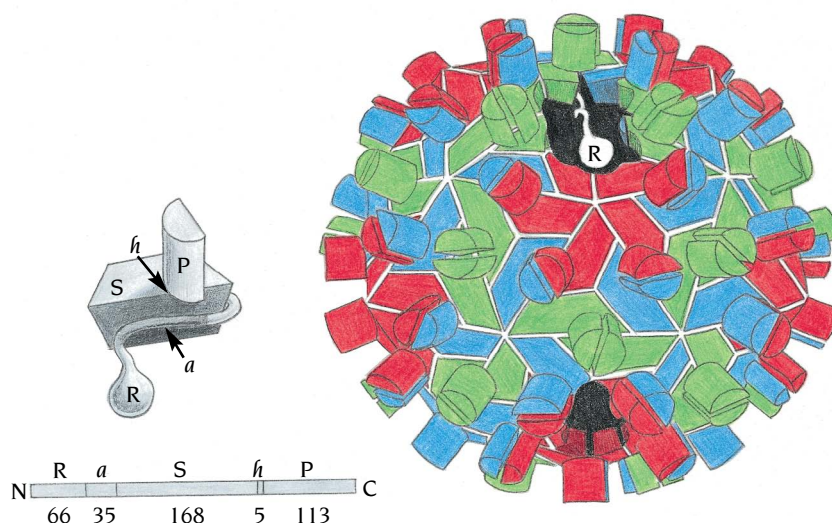


Figure 16.8 Architecture of the tomato bushy stunt virus particle. The polypeptide chain of each subunit folds into three domains (R, S, P) with a 35-residue connecting arm (a) between R and S and a hinge (h) between S and P. The number of amino acid residues in each structural module is indicated. The subunits pack into the virus particle in one of three conformations, colored red (A), blue (B), and green (C). The S domains of the red subunits pack around fivefold axes; the S domains of the blue and green subunits alternate around the threefold axes. The P domains project out from the surface of the particle. The cutaway region in the large diagram shows the connecting arm (a) between domains S and R and the region that the disordered domain R might occupy. (Adapted from S.C. Harrison et al., *Nature* 276: 368–373, 1978.)

quite differently oriented. In addition, the arm region (a) is ordered in C but disordered in A and B. There are no gross conformational differences between A and B, but there are significant local structural differences, especially in the interaction areas. The subunits thus accommodate to the three different environments and preserve quasi-equivalence in most contact regions by conformational differences of the protein chain; structural diversity has been achieved without sacrificing genetic economy.

The S domains form the viral shell by tight interactions in a manner predicted by the Caspar and Klug theory and shown in Figure 16.8. The P domains interact pairwise across the twofold axes and form protrusions on the surface. There are 30 twofold axes with icosahedral symmetry that relate the P domains of C subunits (green) and in addition 60 pseudotwofold axes relating the A (red) and B (blue) subunits (Figure 16.9). By this arrangement the 180 P domains form 90 dimeric protrusions.

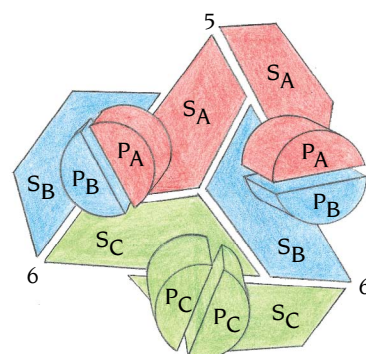
One of the remarkable features of this structure as well as other $T = 3$ plant virus structures is the way in which the ordered connecting arms of C subunits interdigitate to form an internal framework (Figure 16.10). These arms (a in Figure 16.8) extend along the inner face of the S domain and loop around icosahedral threefold axes. Three such C subunit arms contact each other in this way, and all 60 C subunits form a coherent network (see Figure 16.10). The function of this framework is to determine the size of the particle—that is, to ensure that the viral shell closes round on itself correctly during assembly. It does so by serving as a molecular switch. The curvature of the shell along the line joining two threefold axes is much less than the curvature along a line joining a threefold and a fivefold symmetry axis. This difference results from insertion of the ordered C-subunit arms into the C–C contact thereby separating the inner parts of the S domains and flattening the surface lattice.

The size of this viral particle is of course larger than that of a virus with only 60 subunits. The diameter of tomato bushy stunt virus is 330 Å compared with 180 Å for satellite tobacco necrosis virus. The increase in volume of the capsid means that a roughly four times larger RNA molecule can be accommodated.

The protein subunits recognize specific parts of the RNA inside the shell

The N-terminal part of the tomato bushy stunt virus polypeptide chain (the R-segment in Figure 16.8) is disordered in all the subunits. As in the core of many other single-strand RNA viruses this region of the polypeptide chain

Figure 16.9 Contacts between P domains in tomato bushy stunt virus. S_A , S_B , and S_C are the shell domains of subunits A, B, and C, respectively. P_A , P_B , and P_C are the protruding domains of subunits A, B, and C, respectively.



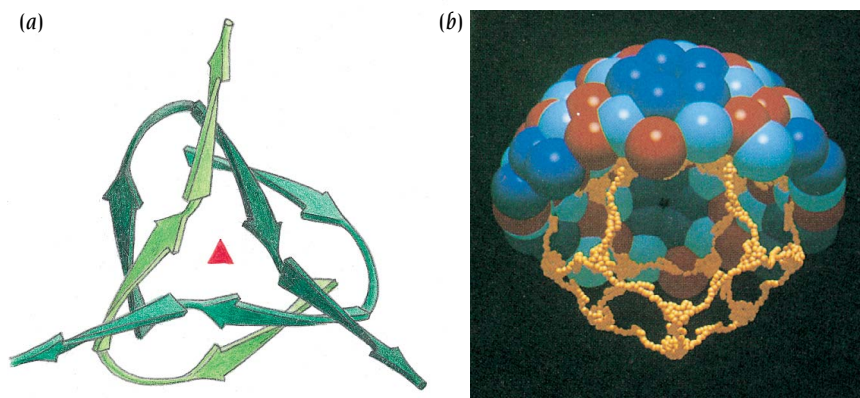


Figure 16.10 The arms of all 60 C subunits in tomato bushy stunt virus form an internal framework. (a) Configuration of interdigitated arms from the three C subunits, viewed down a threefold axis. The β strands are shown as arrows. (b) Cutaway view of the virus particle, emphasizing the framework function of the C-subunit arms. These arms are shown as chains of small balls, one per residue. The region where three arms meet and interdigitate is shown schematically in (a). The main part of each subunit is represented by large balls. Only about one hemisphere of these is drawn, but all the C-subunit arms are included. [(a) Adapted from A.J. Olson et al., *J. Mol. Biol.* 171: 61–93, 1983. (b) Courtesy of A.J. Olson.]

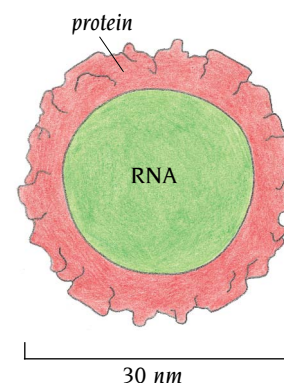
contains many positively charged residues. They appear to serve two functions. One is to bind RNA “nonspecifically” and neutralize its negative charge; the other is to recognize specifically a unique “packaging sequence” in the viral genome. The precise structure of this packaging interaction has yet to be worked out, but it is likely to involve a specific RNA structure and some part of the R-segment bound in a defined way by the RNA.

The protein capsid of picornaviruses contains four polypeptide chains

A protein capsid built up from identical subunits is an economical way for the virus to protect its RNA. However, more elaborate schemes have also evolved, such as that found in **picornaviruses**. These viruses constitute a large family of animal viruses with single-stranded RNA genomes. They have been classified into four groups: (1) **cardioviruses**, such as encephalomyocarditis virus and Mengo virus; (2) **enteroviruses** such as poliovirus and hepatitis A virus; (3) **rhinoviruses**, which cause common colds and of which there are about 100 serotypes; and (4) **aphtoviruses**, which include foot-and-mouth disease virus. The three-dimensional structures of representative examples from each of these groups are known. The group of Michael Rossmann at Purdue University, Indiana, has determined the structure of Mengo virus to 3.0 Å resolution and that of human rhinovirus strain 14 to 2.6 Å resolution. The structure of poliovirus was determined by the group of James Hogle at Scripps Clinic, La Jolla, California, to 2.9 Å resolution, and the structure of foot-and-mouth disease virus, by the group of David Stuart at Oxford University to 2.5 Å resolution.

All these viruses have a molecular mass of around 8.5 million daltons; one long RNA molecule of around 8000 nucleotides accounts for 30% of this molecular mass. The RNA chain occupies the interior of the spherical virion and extends from its center to a radius of about 110 Å (Figure 16.11). A continuous protective shell of protein subunits surrounds the RNA chain. This continuous shell has an average thickness of about 30 Å, giving the virus particle an approximate total diameter of 300 Å. The loops of the polypeptide chain that project from the main part of the protein shell are recognized by the immune defense mechanism of the host, and they contain most of the binding sites for host antibodies. The surface cavities bounded by these protrusions are believed in some cases to be specific sites for receptors on the surface of the host cell. The structural properties of the protein molecules that form the shell must thus include not only the usual requirements for assembly into an icosahedron but also the ability to preserve the receptor sites that render the virus infectious, while accommodating mutations in neighboring antigenic sites in order to escape the immune defenses of the host.

Figure 16.11 Schematic diagram of a picornavirus particle, illustrating the volume occupied by RNA and protein. The surface of the particle contains protrusions and depressions.



There are four different structural proteins in picornaviruses

The shell of all picornaviruses is built up from 60 copies each of four polypeptide chains, called VP1 to VP4. These are translated from the viral RNA into a single polypeptide, which is posttranslationally processed by stepwise proteolysis involving virally encoded enzymes. First, the polypeptide chain is cleaved into three proteins VP0 (which is the precursor for VP2 and VP4), VP1 and VP3. These proteins begin the assembly process. The last step of the processing cascade occurs during completion of the virion assembly; the precursor protein VP0 is cleaved into VP2 and VP4 by a mechanism that is probably autocatalytic but may also involve the viral RNA. VP1, VP2, and VP3 have molecular masses of around 30,000 daltons, whereas VP4 is small, being 7000 daltons, and is completely buried inside the virion.

The arrangement of subunits in the shell of picornaviruses is similar to that of $T = 3$ plant viruses

The capsids of many plant viruses, like tomato bushy stunt virus, have 180 polypeptide subunits ($T = 3$) that are chemically identical. Picornaviruses also have 180 polypeptide subunits in their capsids; these, however, are 60 copies each of three different types, VP1, VP2, and VP3, with no significant amino acid sequence homology between them. The 60 copies of the fourth small subunit, VP4, we consider as a continuation of VP2 that has become detached. Given this diversity of subunits, the picornaviruses could, in principle, form an icosahedral shell with a completely different subunit arrangement and different subunit structures from those of the plant viruses. This is not the case, however. The 180 large subunits of the picornaviruses are arranged in an extraordinarily similar way to the 180 subunits of the $T = 3$ plant viruses. This packing is achieved not by conformational changes, as is the case with the single polypeptide of the plant viruses, but by having within the shell three chemically different polypeptide chains with local structural differences.

The asymmetric unit contains one copy each of the subunits VP1, VP2, VP3, and VP4. VP4 is buried inside the shell and does not reach the surface. The arrangement of VP1, VP2, and VP3 on the surface of the capsid is shown in Figure 16.12a. These three different polypeptide chains build up the virus shell in a way that is analogous to that of the three different conformations A, C, and B of the same polypeptide chain in tomato bushy stunt virus. The viral coat assembles from 12 compact aggregates, or pentamers, which contain five of each of the coat proteins. The contours of the outward-facing surfaces of the subunits give to each pentamer the shape of a molecular mountain: the VP1 subunits, which correspond to the A subunits in $T = 3$ plant viruses, cluster at the peak of the mountain; VP2 and VP3 alternate around the foot; and VP4 provides the foundation. The amino termini of the five VP3 subunits of the pentamer intertwine around the fivefold axis in the interior of the virion to form a β structure that stabilizes the pentamer and in addition interacts with VP4.

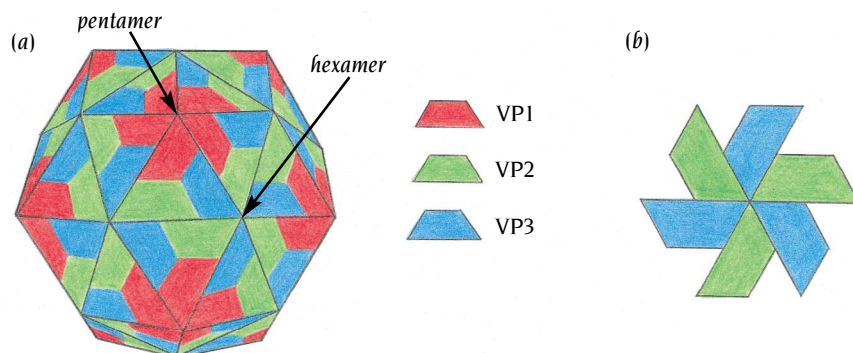


Figure 16.12 (a) Schematic diagram of the surface of a picornavirus. Each triangle represents an asymmetric unit that contains three protein subunits on its surface: VP1 is red, VP2 is green, and VP3 is blue. The diagram illustrates that, at the surface, VP1 subunits are clustered around the fivefold axes, forming pentamers, whereas VP2 and VP3 alternate around the threefold axes, forming hexamers. (b) The hexameric arrangement of VP2 and VP3.

Subunits VP2 and VP3 from different pentamers alternate around the threefold symmetry axes like subunits B and C in the plant viruses (Figure 16.12b). Since VP2 and VP3 are quite different polypeptide chains, they cannot be related to each other by strict symmetry, or even by quasi-symmetry in the original sense of the word. To a first approximation, however, they are related by a quasi-sixfold symmetry axis, since the folded structures of the cores of the subunits are very similar.

The fact that spherical plant viruses and some small single-stranded RNA animal viruses build their icosahedral shells using essentially similar asymmetric units raises the possibility that they have a common evolutionary ancestor. The folding of the main chain in the protein subunits of these viruses supports this notion.

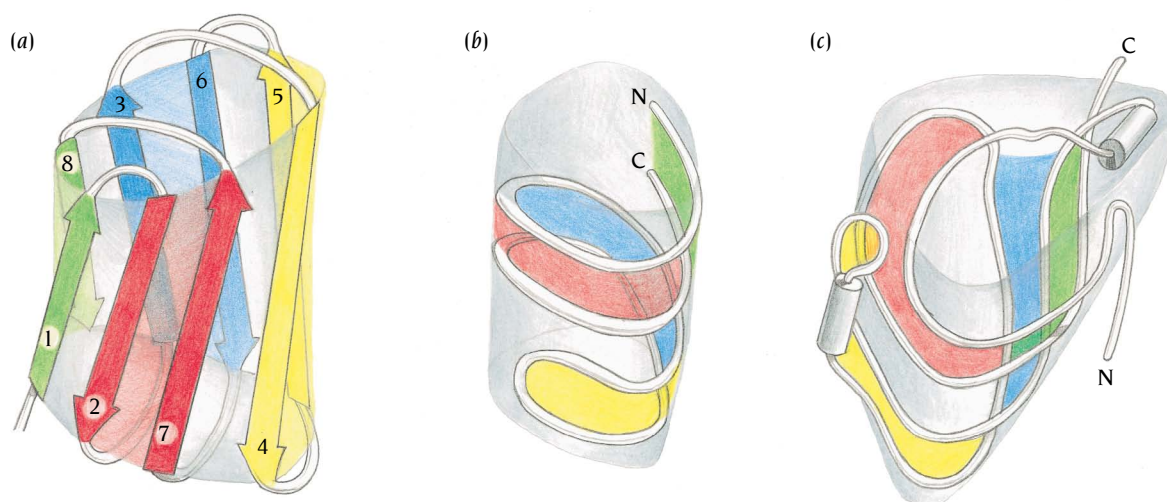
The coat proteins of many different spherical plant and animal viruses have similar jelly roll barrel structures, indicating an evolutionary relationship

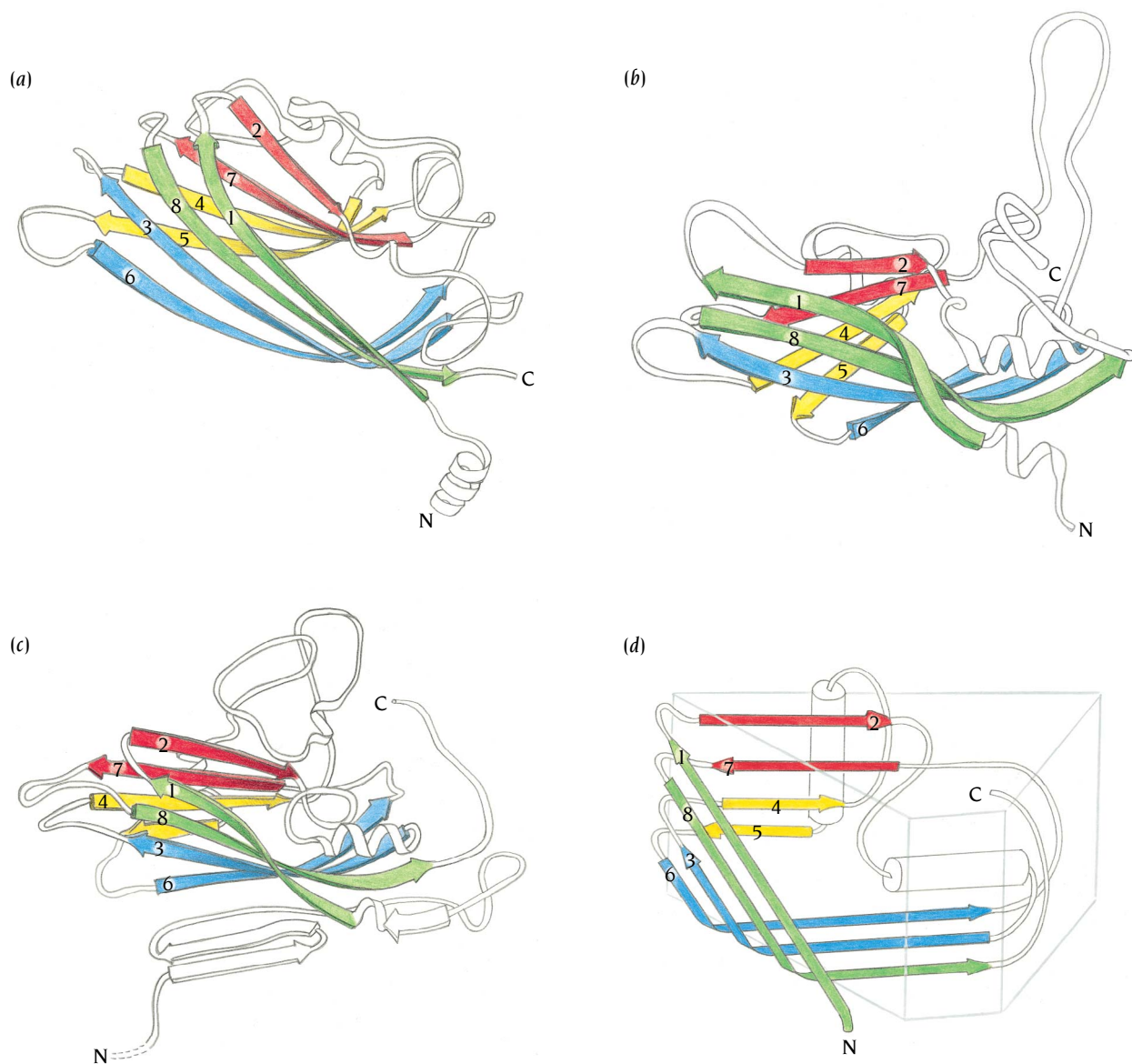
One of the most striking results that has emerged from the high-resolution crystallographic studies of these icosahedral viruses is that their coat proteins have the same basic core structure, that of a jelly roll barrel, which was discussed in Chapter 5. This is true of plant, insect, and mammalian viruses. In the case of the picornaviruses, VP1, VP2, and VP3 all have the same jelly roll structure as the subunits of satellite tobacco necrosis virus, tomato bushy stunt virus, and the other T = 3 plant viruses. Not every spherical virus has subunit structures of the jelly roll type. As we will see, the subunits of the RNA bacteriophage, MS2, and those of alphavirus cores have quite different structures, although they do form regular icosahedral shells.

The canonical jelly roll barrel is schematically illustrated in Figure 16.13. Superposition of the structures of coat proteins from different viruses show that the eight β strands of the jelly roll barrel form a conserved core. This is illustrated in Figure 16.14, which shows structural diagrams of three different coat proteins. These diagrams also show that the β strands are clearly arranged in two sheets of four strands each: β strands 1, 8, 3, and 6 form one sheet and strands 2, 7, 4, and 5 form the second sheet. Hydrophobic residues from these sheets pack inside the barrel.

In all jelly roll barrels the polypeptide chain enters and leaves the barrel at the same end, the base of the barrel. In the viral coat proteins a fairly large number of amino acids at the termini of the polypeptide chain usually lie outside the actual barrel structure. These regions vary considerably both in size and conformation between different coat proteins. In addition, there are three loop regions at this end of the barrel that usually are quite long and that also show considerable variation in size in the plant viruses and the

Figure 16.13 The known subunit structures of plant, insect, and animal viruses are of the jelly roll antiparallel β barrel type, described in Chapter 5. This fold, which is schematically illustrated in two different ways, (a) and (b), forms the core of the S domain of the subunit of tomato bushy stunt virus (c). [(b), (c) Adapted from A.J. Olson et al., *J. Mol. Biol.* 171: 61–93, 1983.]





picornaviruses. In contrast, the four loop regions at the other end of the barrel, the top, are short and exhibit only minor variations in size. As a consequence all coat proteins are shaped like a wedge or flattened cone with a broad base formed by the long loops and a narrow tip where the short loops keep the β strands together (see Figure 16.14d).

The finding that the coat proteins of these spherical viruses have the same general topology and that the subunit arrangements of picornaviruses and T = 3 plant viruses are very similar means it is highly likely that all these viruses are evolutionarily related. Amino acid sequence comparisons reveal relationships only within the same class of viruses. The three major protein chains of a picornavirus, VP1, VP2, and VP3, which range in size from 230 to 280 amino acids, have different sequences and show no obvious homology with each other, either at the gene or protein level. Furthermore, there are no sequence similarities to the coat proteins of spherical viruses of other families such as the plant viruses. In contrast, the same protein from different picornaviruses show significant sequence similarity. For example, VP2 from Mengo virus and rhinovirus have about 30% sequence identity. Similar sequence similarity is found in other proteins that are coded by the

Figure 16.14 Schematic diagrams of three different viral coat proteins, viewed in approximately the same direction. Beta strands 1 through 8 form the common jelly roll barrel core. (a) Satellite tobacco necrosis virus coat protein. (b) Subunit VP1 from poliovirus. (c) Subunit VP2 from human rhinovirus. (d) Canonical jelly roll barrel, illustrating the wedge-shaped arrangement of the β strands in the subunit structures of viruses. [(a) Adapted from L. Liljas et al., *J. Mol. Biol.* 159: 93–108, 1982. (b) Adapted from J. Hogle et al., *Science* 229: 1358–1365, 1985. (c) Adapted from M.G. Rossmann et al., *Nature* 317: 145–153, 1985.]

picornaviral genome, such as RNA polymerase. The picornaviruses obviously are an evolutionarily related family.

From sequence data alone no one would suggest that picornaviruses and $T = 3$ plant viruses are related. However, in order to obtain both the same topology of the subunits and a similar packing arrangement, there must be constraints on a large number of amino acids. Even though they are not the same in the different proteins, they must fulfill the same functional role. Despite their different sequences, the common topology and packing properties suggest that they all derive from the same ancestor. The alternative, that both the same topology and packing arrangements of the coat proteins have arisen independently many times during evolution, seems much less likely.

Drugs against the common cold may be designed from the structure of rhinovirus

Most drugs are effective because they bind to a specific receptor site and block the physiological function of a protein. Classical drug design has been based on this concept for several decades. Compounds that are structural variants of substrates for selected target proteins are synthesized and tested in binding studies. In this way it is usually possible to obtain inhibitors that bind better than the physiological substrates, although several thousands of compounds may have to be synthesized and tested before a suitable drug is found. It has, therefore, been the dream of many pharmaceutical chemists to be able to design new drugs in a more rational way based on a knowledge of the three-dimensional structure of the binding sites of relevant proteins. Understandably, Michael Rossmann attracted considerable attention in 1986 by reporting a detailed crystallographic analysis of the binding of an antiviral drug called WIN 51711 to **human rhinovirus**. This work raised hopes that it might eventually be possible to design efficient drugs against the common cold, as well as many other viral diseases.

The cleft where this drug binds is inside the jelly roll barrel of subunit VP1. Most spherical viruses of known structure have the tip of one type of subunit close to the fivefold symmetry axes (Figure 16.15a). In all the picornaviruses this position is, as we have described, occupied by the VP1 subunit. Two of the four loop regions at the tip are considerably longer in VP1 than in the other viral coat proteins. These long loops at the tips of VP1 subunits protrude from the surface of the virus shell around its 12 fivefold axes (Figure 16.15b).

In rhinoviruses there are depressions, or “canyons,” which are 25 Å deep and 12 to 30 Å wide and which encircle the protrusions (Figure 16.15b). One wall of the canyons is lined by residues from the base of VP1. The structure of VP1 is such that the barrel is open at the base and permits access to the hydrophobic interior of the barrel, as in the up-and-down barrel structure of the retinol-binding protein described in Chapter 5.

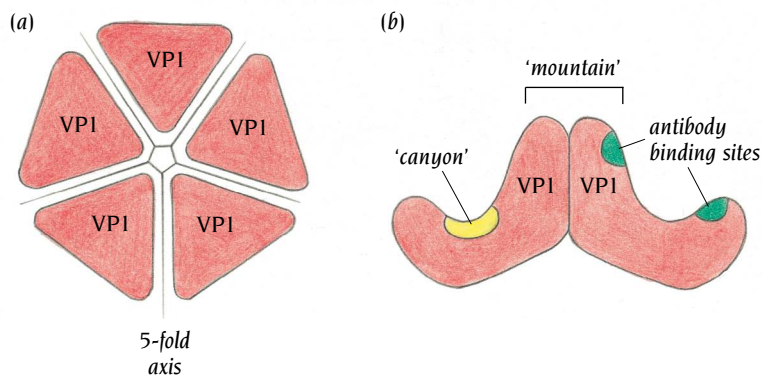


Figure 16.15 Subunits VP1 in picornaviruses cluster around the fivefold axis (a) so that their tips form a protrusion on the viral surface. In rhinoviruses a depression or “canyon” surrounds the protrusion and contains a binding site for initial attachment of the virus to its receptor, the adhesion molecule ICAM-1. (b) Antiviral agents bind below the floor of this canyon in a cavity within VP1. [(b) Adapted from M. Luo et al., *Science* 235: 182–191, 1987.]

The antiviral compound that Rossmann studied binds inside the barrel of VP1 (Figure 16.16a). It appears to gain access to this site through an opening of the floor of the canyon. One end of the compound is a 3-methylisoxazole group that is inserted into the hydrophobic interior of the VP1 barrel (Figure 16.16b). This end is connected by an aliphatic chain to a 4-oxazolinyloxy group, OP. Both the aliphatic chain and the OP group are contacted by side chains of the amino acids that line the bottom of the canyon. It is quite apparent by looking at the geometries of the cleft and the antiviral compound that the fit is not optimal and that it should be possible to design compounds with stronger binding to this site. In fact, such compounds have been designed and synthesized and are now subject to clinical trials.

Rossmann suggested that the canyons form the binding site for the rhinovirus receptor on the surface of the host cells. The receptor for the major group of rhinoviruses is an adhesion protein known as ICAM-1. Cryoelectron microscopic studies have since shown that ICAM-1 indeed binds at the canyon site. Such electron micrographs of single virus particles have a low resolution and details are not visible. However, it is possible to model components, whose structure is known to high resolution, into the electron microscope pictures and in this way obtain rather detailed information, an approach pioneered in studies of muscle proteins as described in Chapter 14.

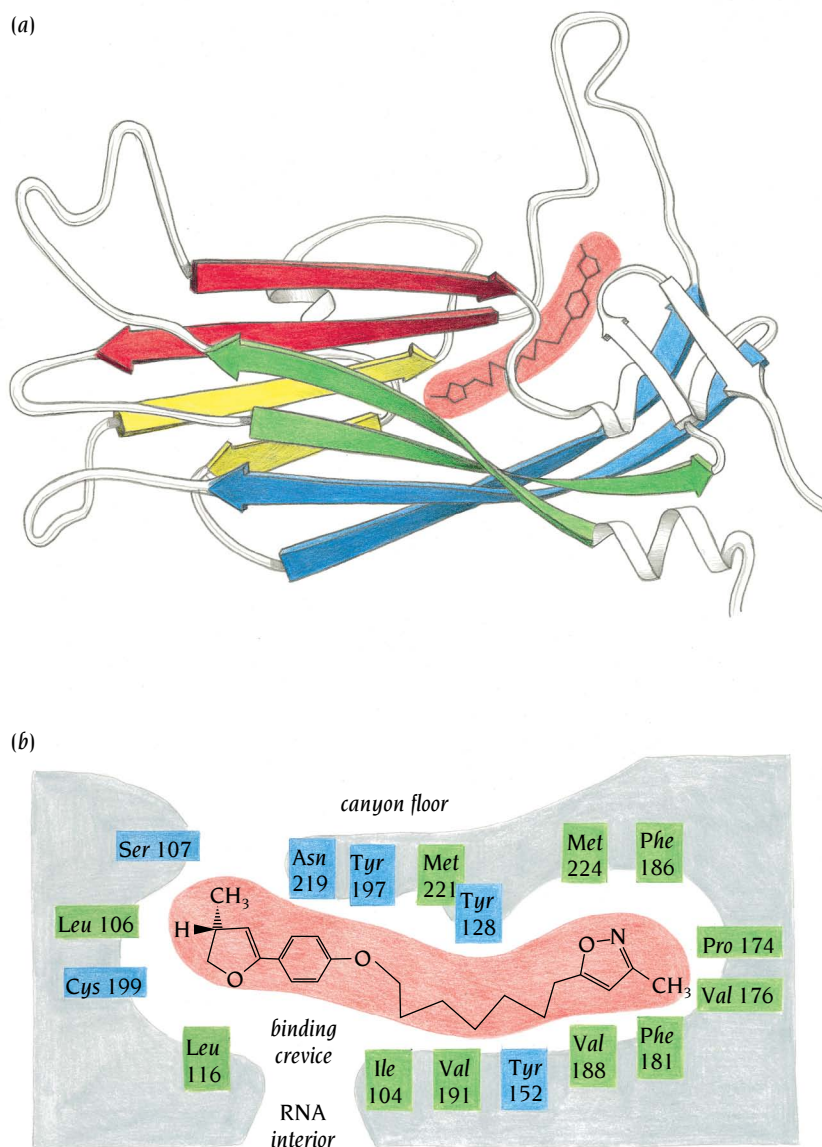


Figure 16.16 Schematic diagrams illustrating the binding of an antiviral agent to human rhinovirus strain 14. (a) The drug binds in a hydrophobic pocket of VP1 below the floor of the canyon. (b) Schematic diagram of VP1 illustrating the pocket in the jelly roll barrel where the drug binds. (Adapted from T.J. Smith et al., *Science* 233: 1286–1293, 1986.)

Bacteriophage MS2 has a different subunit structure

The structures of many different plant, insect, and animal spherical viruses have now been determined to high resolution, and in most of them the subunit structures have the same jelly roll topology. However, a very different fold of the subunit was found in **bacteriophage MS2**, whose structure was determined to 3 Å resolution by Karin Valegård in the laboratory of Lars Liljas, Uppsala.

MS2 belongs to a family of small single-stranded RNA phages—MS2, R17, f2, and Q β —that infect *Escherichia coli*. These phages are about 250 Å in diameter and contain only four genes in their 4000-nucleotide-long chromosome. One of these genes codes for the 129 amino acid coat protein that is present in 180 copies in the T = 3 virus particle. A second gene encodes a protein 393 amino acids long that is present in a single copy in each virus particle and is needed both for attachment of the phage to its host and penetration of the viral RNA into the bacterium.

The x-ray structure of the MS2 particle was determined by similar methods to those applied to other viral structures; the icosahedral symmetry of the particle was used to compute an average electron density for the subunits. The electron densities for both the attachment protein and the RNA, one copy of which are present in each virus, are lost in this process and only the structure of the coat protein is obtained. The subunit folds into a five-stranded up-and-down antiparallel β sheet with an additional short hairpin at the amino terminus and two α helices at the other end of the chain (Figure 16.17). The α helices are responsible for interactions with a second subunit to form a tight dimer. The two β sheets, one from each subunit in the dimer, are aligned at their edges so that a large continuous β sheet of 10 adjacent antiparallel β strands is formed, with α helices from one subunit packing against the β sheets of the second subunit (Figure 16.18). This structure is quite different from the jelly roll found in all other spherical viruses so far.

Since all members of this family of RNA phages have homologous coat proteins, their subunits are expected to have the same three-dimensional structure. It remains to be seen if the MS2 fold is also present in any other unrelated viruses. The fold is so far unique for the MS2 subunit, but similar structures have been observed in other proteins such as the major histocompatibility antigen, HLA, which was discussed in Chapter 15.

A dimer of MS2 subunits recognizes an RNA packaging signal

The coat protein of MS2 recognizes and binds to a sequence of 19 nucleotides in the viral genome that contains the start codon for the phage replicase gene. This recognition controls two processes: translation of the replicase protein and packaging of the correct RNA molecule, the phage genome, into the protein shell. In solution, the MS2 subunits associate as stable dimers, and encapsidation of viral genomic RNA is initiated when a dimer binds to the 19-nucleotide recognition sequence. Once this assembly initiation complex has been formed, capsid growth occurs, with coat protein dimers adding to the complex until the complete virus particle has been formed.

Capsids can also be formed in the absence of RNA as well as in the presence of small RNA molecules comprising the 19 nucleotides of the

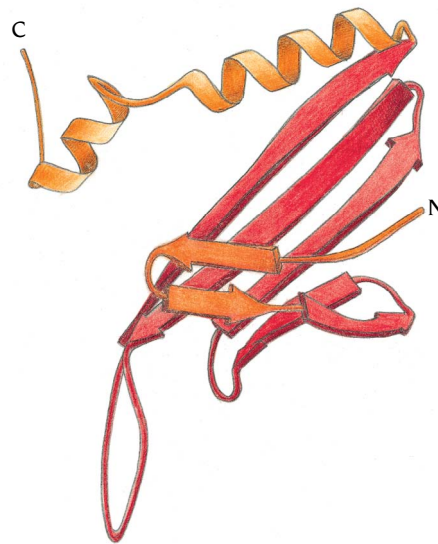


Figure 16.17 The subunit structure of the bacteriophage MS2 coat protein is different from those of other spherical viruses. The 129 amino acid polypeptide chain is folded into an up-and-down antiparallel β sheet of five strands, β 3– β 7, with a hairpin at the amino end and two C-terminal α helices. (Adapted from a diagram provided by L. Liljas.)

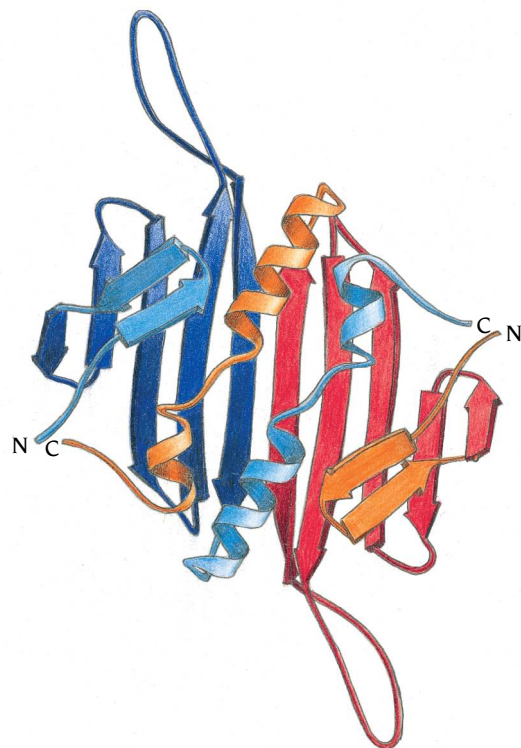


Figure 16.18 A dimer is the basic unit that builds up the capsid of bacteriophage MS2. The two subunits (red and blue) are arranged so that the dimer has a β sheet of 10 antiparallel strands on one side and the hairpins and α helices on the other side. The helices from one subunit pack against β strands from the other subunit and vice versa. (Adapted from a diagram provided by L. Liljas.)

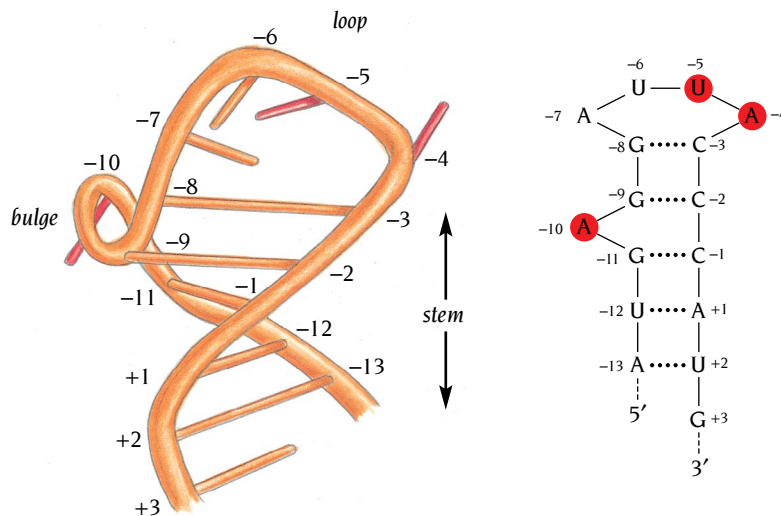


Figure 16.19 Schematic drawing illustrating the structure and sequence of the RNA fragment that is recognized and bound by the coat protein of bacteriophage MS2. The RNA fragment forms a base-paired stem with a bulge at base -10 and a loop of four bases. Bases that form sequence-specific interactions with the coat protein are red. (Adapted from a diagram provided by L. Liljas.)

recognition sequence. The structure of such a capsid complex has been determined by the group of Lars Liljas, and it has given insight into the recognition of viral RNA by the MS2 coat protein. The RNA molecule forms a stem of five base pairs with a loop of four nucleotides at the top (Figure 16.19). There is a bulge in the middle of the stem causing an adenine base to project out from the base-paired double helix of the stem. This adenine in combination with two bases in the loop region form sequence-specific interactions with amino acid residues from the extended β sheet of the coat protein dimer (Figure 16.20). These interactions require a complementarity between the structures of the RNA and of the protein. Consequently, recognition involves not only the proper sequence of the three interacting bases in the RNA but also the formation of a stem-loop structure, as well as a rigidly folded and assembled protein dimer.

The core protein of alphavirus has a chymotrypsin-like fold

Alphaviruses, such as Sindbis virus and Semliki Forest virus, are a group of mosquito-borne, enveloped RNA viruses that can cause encephalitis, fever, arthritis and rashes in mammals. These viruses have two protein shells—an outer glycoprotein layer and an inner core—which are separated by a lipid bilayer, a membrane. Studies by cryoelectron microscopy have shown that

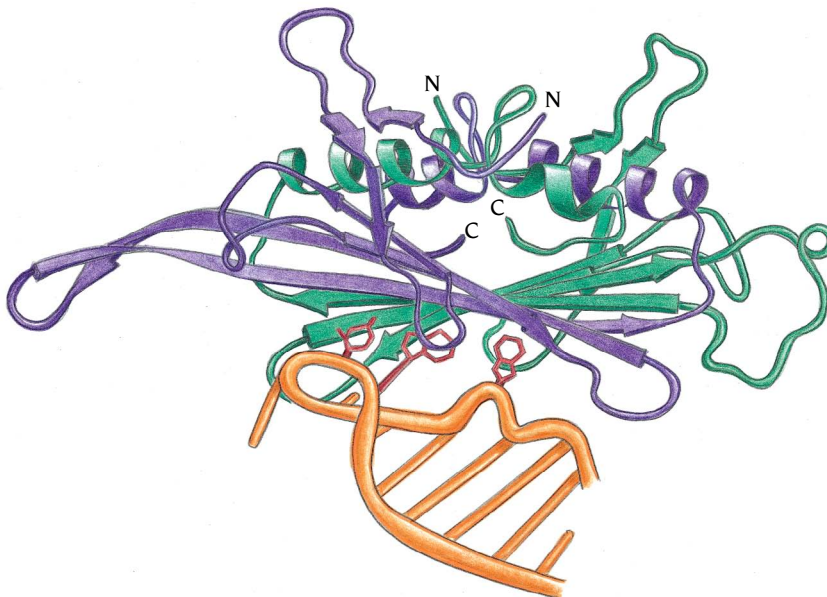


Figure 16.20 The structure of the complex between a dimer of the coat protein of bacteriophage MS2 and the RNA fragment shown in Figure 16.19. One subunit of the coat protein dimer is green, the other is violet and the RNA fragment is orange. Bases that form sequence specific interactions with the protein are red. (Adapted from a diagram provided by L. Liljas.)

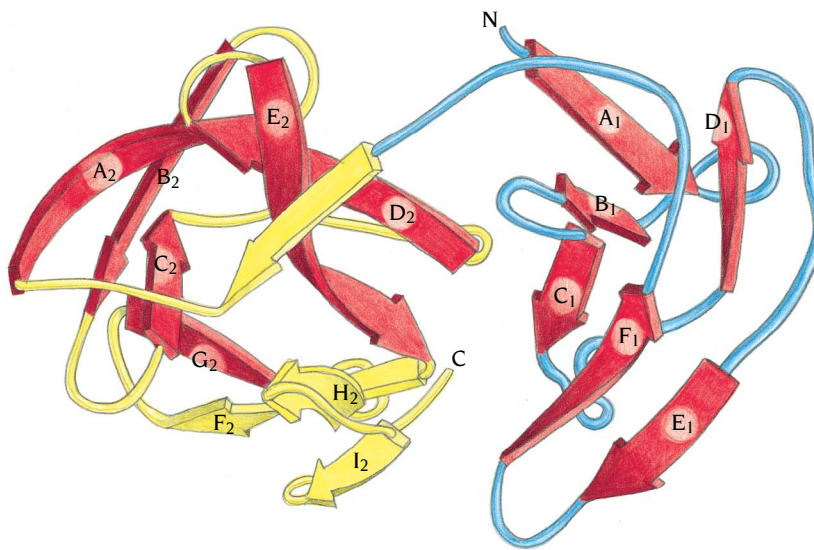


Figure 16.21 Structure of one subunit of the core protein of Sindbis virus. The protein has a similar fold to chymotrypsin and other serine proteases, comprising two Greek key motifs separated by an active site cleft. The C-terminus of the protein is bound in the catalytic site, making the coat protein inactive. (Adapted from S. Lee et al., *Structure* 4: 531–541, 1996.)

both the core and the outer glycoprotein shell conform to $T = 4$ icosahedral lattices. No high-resolution structure of an alphavirus has yet been obtained, but the subunit structure of the core protein of **Sindbis virus** has been determined by x-ray crystallography by the group of Michael Rossmann. It differs from both the MS2 coat protein and the jelly roll fold.

The core protein of alphaviruses is generated from a longer precursor protein by an autocatalytic cleavage; in other words the precursor protein has protease activity. This activity resides in the part of the precursor that after cleavage forms the inner shell of the virus, the core. Unassembled subunits of this core protein have two domains: a flexible N-terminal domain of roughly 110 residues rich in positive charges, which binds to RNA, and a 150-residue C-terminal domain with a structure similar to chymotrypsin and other serine proteases described in Chapter 11. In Rossmann's crystals the flexible N-terminal domain is not visible. The C-terminal domain comprises two Greek key β barrel motifs with the active site in a cleft between them (compare Figures 16.21 and 11.7). In contrast to chymotrypsin, there is only one polypeptide chain, and there are no disulfide bridges. Like chymotrypsin, however, the coat protein protease domain has a catalytic triad of Asp-His-Ser residues in the active site, and therefore the catalytic mechanism of cleavage is similar to chymotrypsin (see Chapter 11).

Each precursor protein molecule is cleaved only once to generate one molecule of the coat protein, and catalytic activity is restricted to the precursor protein. Why is the coat protein itself catalytically inactive? The structure of the coat protein shows that its C-terminus is bound in the active site cleft and thereby prevents other proteins entering the cleft and being cleaved. This arrangement allows the precursor protein to fulfill its function to generate the coat protein and prevents the coat protein from destroying other proteins in the infected cell, including other coat proteins.

SV40 and polyomavirus shells are constructed from pentamers of the major coat protein with nonequivalent packing but largely equivalent interactions

Polyomavirus, simian virus 40 (SV40), and the papilloma viruses are members of the **papovavirus** family and were for a long time thought to have a simple $T = 7$ icosahedral design. Electron micrographs by Klug and co-workers of negatively stained particles showed strongly contrasted projections, or capsomers, that would correspond to clusters of five and six subunits around the 12 fivefold and the 60 pseudosixfold positions of an icosahedral $T = 7$ lattice. This arrangement would give 72 capsomers and 420 subunits ($7 \times 60 = 12 \times 5 + 60 \times 6$), per virus particle.

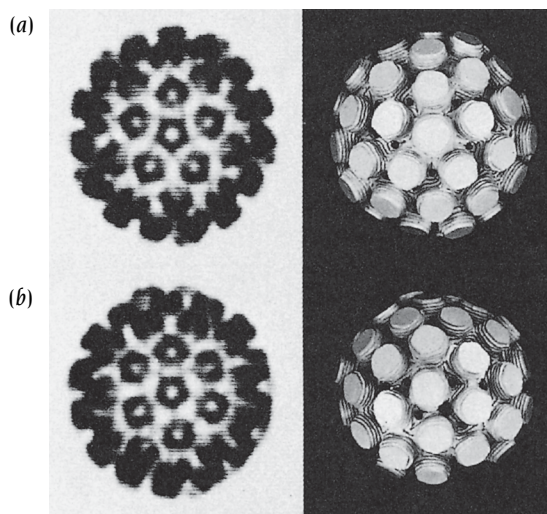


Figure 16.22 Polyomavirus was believed from electron micrographs to have a simple $T = 7$ design until x-ray studies showed otherwise. A view down the fivefold symmetry axis of the icosahedral structure (a) shows that the central capsomer is pentameric in shape and surrounded by five other capsomers as expected. The view down the pseudosixfold axis (b) shows, however, that the central capsomer is pentameric in shape and not hexameric as required for a $T = 7$ structure. (Adapted from I. Rayment et al., *Nature* 295: 110–115, 1982, by copyright permission of Macmillan Magazines Limited.)

X-ray studies at 22.5 Å resolution of murine **polyomavirus** by I. Rayment and D.L.D. Caspar at Brandeis University confirmed the presence of these 72 capsomers at the expected positions, but even at low resolution the pentagonal shape of all 72 capsomers was evident (Figure 16.22). They concluded that each capsomer must be a pentameric assembly of the major viral subunit, known as viral protein 1 (VP1). Each of the 60 icosahedral asymmetric units contains 6 VP1 subunits, not 7, and the complete shell contains 360 VP1 subunits. The 12 VP1 pentamers centered on icosahedral fivefold axes are identically related to their five neighbors, but the 60 pentamers centered on pseudosixfold positions “see” each of their 6 neighbors quite differently (Figure 16.23). How can such diversity of interaction be incorporated into the bonding properties of just one type of protein subunit, without compromising specificity and accuracy of assembly?

The structures of **SV40** and polyomavirus, determined by Stephen Harrison and coworkers at 3.1 Å and 3.6 Å resolution, respectively, provide a striking answer to this question. The major part of each VP1 subunit folds into a jelly roll domain, which together with four others make up the core of the pentameric capsomers (Figures 16.24 and 16.25). About 45 N-terminal residues and 60 C-terminal residues are not part of the jelly-roll domain. The C-terminal segment projects from each VP1 subunit and “invades” a neighboring pentamer. Part of this C-terminal arm forms a β strand, which augments one β sheet of the jelly roll in a VP1 subunit in the invaded pentamer (yellow in Figure 16.25). This β strand is clamped in place by an additional short strand made by residues in the N-terminal segment of the target subunit (see Figure 16.25). Thus, the pentamers are building blocks tied together by the C-terminal arms of their subunits rather than cemented together across preformed complementary interfaces (Figure 16.26).

The interactions of all 360 arms with their targets are essentially identical. Most of the variability is in how the arms span the interpentamer gap—that is in how they get from their subunit of origin to the pentamer they invade. By expanding our notion of functional contacts between subunits to include such linkages, rather than restricting ourselves to the extensive surface contacts implied by diagrams such as those in Figures 16.6 or 16.7, we can readily explain the high degree of specificity seen in these assemblies.

The links between VP1 pentamers are quite tenuous. With one exception, (across the twofold axis between the γ subunits in Figure 16.26), all

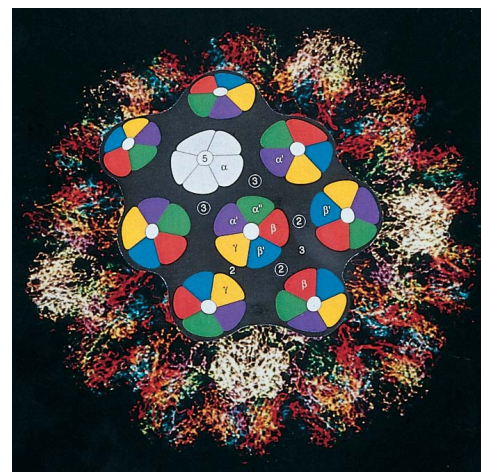


Figure 16.23 Overview of the structure of the SV40 virus particle, showing the packing of pentamers. The subunits of pentamers on fivefold positions are shown in white; those of pentamers in six-coordinated positions are shown in colors. The six colors indicate six quite different environments for the subunit. (Courtesy of S. Harrison.)

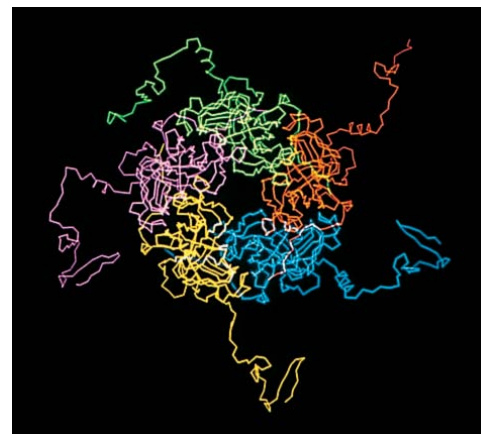


Figure 16.24 A six-coordinated pentamer of SV40 “extracted” from the model shown in Figure 16.23. The extended carboxy-terminal arms are shown in the conformations they adopt in the assembled particle; in the free pentamer they are disordered and flexible. (Courtesy of S. Harrison.)

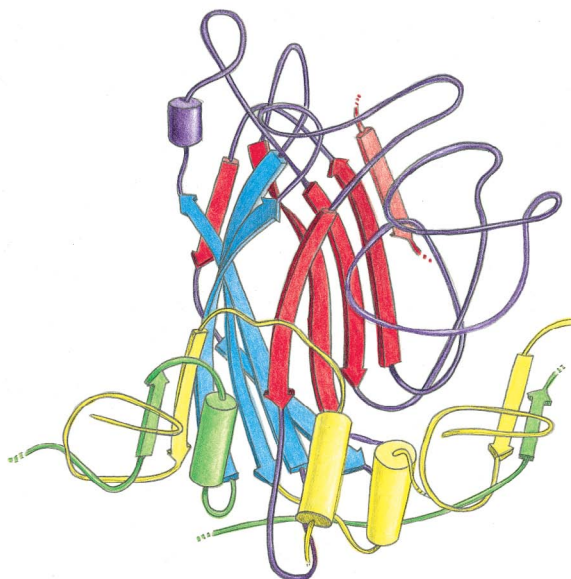


Figure 16.25 Schematic diagram of the SV40 subunit, β strands are represented as ribbons; α helices as cylinders and loops as narrow tubes. The two sheets of the β barrel domain are colored blue and red, the intervening loops are violet; the carboxy-terminal arms are yellow and the amino-terminal arms are green. The β strand in pale red is from a neighboring subunit in the pentamer. The complete yellow carboxy-terminal arm emanates from a subunit in another pentamer; only the initial, α -helical segment of the arm from this subunit is shown, since it extends out of the page. Note how the green amino-terminal segment clamps the invading yellow arm in place. (Courtesy of S. Harrison.)

interpentamer contacts involve the C-terminal arms—a single segment of polypeptide chain rather than a complex surface. Rigidity is imparted to the shell by short α helices that are formed by the proximal segments of the C-terminal arms (see Figure 16.26). These helices act like wedges, stabilizing the curvature of the capsid. We can think of the C-terminal arms as ropes that tie together the pentameric building blocks. We can then extend the analogy by imagining that the ropes are drawn tight and that these helices act like small, rigid spacers, which bear against each other and against the pentamer cores.

The augmentation of a β sheet in one protein by a strand emanating from another is a mode of protein association not restricted to viral shells. Small domains involved in intracellular signal transduction bind to “arms” of other proteins by presenting the edge of a sheet on which those arms can form an additional strand.

Quasi-equivalence was conceived to explain why icosahedral symmetry should be selected for the design of closed containers built of a large number of identical structural units with conserved binding specificity. In simple viruses preformed complementary surfaces provide quasi-equivalent interactions between the units. In the papovaviruses on the other hand, the pentameric units are packed in a nonequivalent way and linked together by extended polypeptide chains that become ordered only on association with targets. The actual contacts between different pentamers are largely equivalent, differing primarily by the conformations of the linker regions.

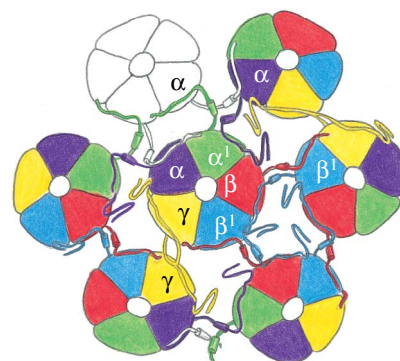
Conclusion

Small spherical viruses have a protein shell around their nucleic acid that is constructed according to icosahedral symmetry. Objects with icosahedral symmetry have 60 identical units related by fivefold, threefold, and twofold symmetry axes. Each such unit can accommodate one or several polypeptide chains. Hence, virus shells are built up from multiples of 60 polypeptide chains. To preserve quasi-equivalent symmetry when packing subunits into the shell, only certain multiples ($T = 1, 3, 4, 7, \dots$) are allowed.

Satellite tobacco necrosis virus is an example of a $T = 1$ virus structure. The 60 identical subunits interact tightly around the fivefold axes on the surface of the shell and around the threefold axes on the inside. These interactions form a scaffold that links all subunits together to complete the shell.

Tomato bushy stunt virus is a $T = 3$ plant virus with 180 chemically identical subunits. Each polypeptide chain is divided into several domains. The subunits preserve quasi-equivalent packing in most contact regions by conformational differences of the protein chains, especially a large change in

Figure 16.26 Schematic diagram illustrating the different conformations of the flexible arms that link the pentamers in SV40. The color scheme corresponds to Figure 16.23. The body of a pentamer is shown as a five-petaled flower; the arms are cylinders (α helices) and lines.



the orientation of the domains. A 35-residue region in 60 of the subunits forms a connected network of interactions around the icosahedral threefold axis that determines the size and shape of the particle.

Picornaviruses construct their shells from 60 copies each of three different polypeptide chains. These 180 subunits are arranged within the shell in a manner very similar to the 180 identical subunits of bushy stunt virus. In some picornaviruses there are protrusions around the fivefold axes, which are surrounded by deep “canyons.” In rhinoviruses, the canyons form the virus’s attachment site for protein receptors on the surface of the host cells, and they are adjacent to cavities that bind antiviral drugs.

Most subunits of plant and animal spherical viruses have the same topology in their core, that of a jelly roll antiparallel β barrel structure. This implies evolutionary relationships even though there is no significant sequence similarity. The barrel is flattened with short loop regions at the top and long at the base, giving the barrel a wedge shape. The subunit structures of alphaviruses and of the bacteriophage MS2 have different arrangements of antiparallel β strands. The coat protein of the core of Sindbis virus has a chymotrypsin fold and its precursor protein has an autocatalytic protease activity. Dimers of the MS2 coat protein bind to a specific recognition structure of RNA, initiating assembly of the bacteriophage particle.

The capsids of polyoma virus and the related SV40 have icosahedral symmetry, with 72 pentameric assemblies of the major capsid protein. The pentamers are linked to their neighbors by flexible arms, with a β strand that augments a β sheet in the invaded pentamer. These flexible arms allow the pentamers to be linked together with both fivefold and sixfold symmetry.

Selected readings

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